

Distributed Machine Learning for Autonomous Agent Swarm: A Survey

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Abstract—Autonomous agents, including unmanned aerial vehicles (UAVs), unmanned ground vehicles (UGVs), unmanned surface vessels (USVs), and unmanned underwater vehicles (UUVs), are widely applied across diverse fields, such as environmental monitoring, logistics, exploration, and military applications, due to their ability to operate autonomously in complex environments. As tasks grow in complexity, there is a shift from individual autonomous agents to collaborative autonomous agent swarms (AASs). These swarms, connected through wireless links, leverage collective intelligence to perform more sophisticated tasks than individual autonomous agents can achieve alone. Deploying AASs introduces challenges in coordination, resource utilization, and adaptability, which traditional model-based methods struggle to address. Recent advancements in machine learning (ML), particularly distributed machine learning (DML) techniques such as federated learning (FL) and multi-agent reinforcement learning (MARL), offer promising solutions. These techniques enable autonomous agents to learn collaboratively without centralizing data, thereby preserving efficiency while adapting to dynamic environments. However, applying DML in AASs presents unique challenges due to autonomous agents' dynamic movement, unreliable communication links, harsh operating conditions, heterogeneity, and resource constraints. Therefore, this survey provides a comprehensive review of integrating DML into AASs. Specifically, we analyze

four representative types of autonomous agents in a unified perspective examining their fundamental characteristics, communication models, and dynamic behaviors. Then, we discuss the limitations of basic ML models in AASs, highlighting the need for DML and the key requirements and metrics for its successful implementation. Furthermore, we explore recent advancements in FL and MARL as applied to AASs, highlighting use cases and key techniques. By identifying current technological progress and gaps in the literature, this survey offers valuable insights for researchers and practitioners and outlines potential directions for future research to enhance the capabilities of AASs through DML.

Index Terms—Autonomous agent swarm, distributed machine learning, swarm intelligence, federated learning, multi-agent reinforcement learning.

I. INTRODUCTION

IN THE last few decades, autonomous agents have emerged as a focal point of research and industrial applications. Autonomous agents are designed to operate in dynamic and unpredictable environments. They make real-time decisions based on sensor inputs without explicit instructions. This category encompasses a diverse array of entities, including but not limited to unmanned aerial vehicles (UAVs), unmanned ground vehicles (UGVs), unmanned surface vessels (USVs), unmanned underwater vehicles (UUVs), and so forth. Each of them is equipped with capabilities to assess situations, plan actions, and execute tasks autonomously.

While individual autonomous agents have demonstrated considerable utility across various applications, the escalating complexity and scale of advanced tasks necessitate a paradigm shift towards collaborative endeavours. This has led to the emergence of the autonomous agent swarm (AAS), a concept where multiple autonomous agents work collaboratively, as shown in Fig. 1. They employ collective intelligence, facilitated by wireless connections that enable information exchange and collaboration, to perform tasks more complex than those possible by individual agents. This collaboration is not merely a summation of individual efforts; it represents a synergistic interaction where the swarm's collective intelligence and capabilities surpass those of its individual components. Early studies have focused on communication and networking [1], [2], [3], [4], [5], [6], cooperative localization [7], [8], [9], cooperative perception [10], [11], [12], [13], [14], formation control and path planning [15], [16], [17], [18], [19], and task assignment [20], [21], [22], [23], [24], [25]. However, the deployment of AAS still faces

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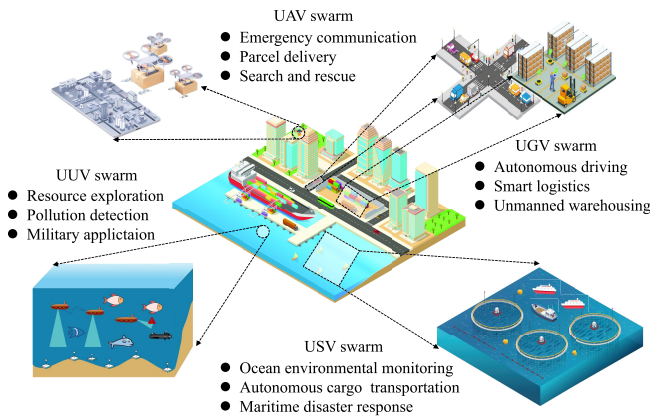


Fig. 1. Application of various AASs.

numerous challenges. Orchestrating a swarm necessitates sophisticated coordination mechanisms to ensure effective collaboration among autonomous agents while avoiding conflicts and redundancies. Meanwhile, due to the limited resources of autonomous agents, this level of cooperation must be achieved without compromising resource utilization efficiency. This includes considerations like the number of agents used, energy consumption, and the usage of communication and computation resources. Conventional methodologies, predominantly reliant on predefined models and heuristic algorithms, struggle to address the dynamic intricacies associated with AAS. These methods often exhibit significant limitations in flexibility and scalability, thus rendering them inadequate for managing the complex interactions and evolving challenges inherent to swarm operations [26]. Therefore, there is a need for novel methodologies to advance real-world applications.

Recently, machine learning (ML) has garnered widespread attention, especially with the advancements in deep learning (DL) and reinforcement learning (RL) over the last decade. These developments present revolutionary opportunities for enhancement across numerous technological domains, such as computer vision, natural language processing, and autonomous driving. These advancements underscore the transformative potential of ML in augmenting the operational capabilities of AASs within a data-driven paradigm, such as object recognition, trajectory planning, simultaneous localization and mapping (SLAM), and so forth. By leveraging data-driven insights and adaptive algorithms, ML enables these swarms to observe complex environments more clearly and make smarter decisions, thereby amplifying their swarm intelligence and efficiency in real-world applications [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43].

Therefore, it has become a critical issue to leverage the data scattered across numerous autonomous agents to train powerful, robust, and efficient ML models. Clearly, it is impractical for autonomous agents to independently train their ML models, relying solely on the limited number of data samples they independently acquire. While centralized ML, particularly cloud-based approaches, has shown remarkable success in recent years, it proves inadequate for AASs. Specifically, the challenges faced by centralized ML in AASs can be categorized into two main types, which fundamentally

arise from the two types of ML models deployed on the autonomous agents in AASs:

- **Independent Tasks:** Tasks executed autonomously by individual agents, such as target observation and individual dynamic control.
- **Collaborative-Competitive Tasks:** Tasks requiring collaboration or competition among autonomous agents, such as communication, networking, formation control, and mission planning.

The challenges faced by centralized ML in AASs can be summarized as follows:

- **Challenges for Independent Tasks:** In AASs, each autonomous agent independently observes or generates data, resulting in inherent data dispersion. While agents may share similar tasks or objectives, individual agents often lack sufficient data to train a high-precision model. The distributed nature of the data makes centralized ML inefficient, increasing bandwidth and computational costs.
- **Challenges for Collaborative-Competitive Tasks:** These tasks involve the interdependence of agents' positions, states, and decisions. Real-time model updates are required to adapt to dynamic environments. Centralized ML struggles to synchronize all the data and make decisions within a short time frame, resulting in delays and inefficiencies that fail to meet the stringent real-time performance and efficiency requirements of AASs.

Distributed machine learning (DML), represented by federated learning (FL) and multi-agent reinforcement learning (MARL), can effectively address the aforementioned issues. FL is an efficient DML framework that focuses on distributed data processing across multiple autonomous agents, enabling them to independently train models and aggregate model updates without centralizing data. This approach significantly reduces communication overhead while maintaining some level of security by ensuring that sensitive data does not need to be shared. In contrast, MARL involves multiple autonomous agents learning optimal strategies within a shared environment, where each agent improves its decision-making through interactions with the environment and other agents. Several surveys have discussed the use of DML in various fields, such as wireless networks [44], [45], [46], [47], [48], [49] and the Internet of Things (IoT) [50], [51], [52], [53]. However, there is currently no comprehensive survey on the application of DML to AASs. Due to the specific characteristics of AASs, employing DML in AASs presents unique challenges that distinguish it from the aforementioned applications.

- **Dynamic Movement.** The primary characteristic of AASs is the mobility of their agents. Depending on the type, the speed of autonomous agents can range from several kilometers per hour to hundreds of kilometers per hour, resulting in a highly dynamic and unstable network topology within AASs.
- **Non-Ideal Environment.** Unlike stable working environments of the Internet or IoT, the working environments of AASs are often harsh. Examples include UAV swarms in forest fire rescues and UUV and USV swarms in maritime

search and rescue operations. These challenging environments not only compromise the reliability of individual agents but also the communication links between them.

- **Heterogeneity.** Autonomous agents within an AAS are often heterogeneous. This heterogeneity may be in type, such as air-ground swarm composed of UAVs and UGVs, or may arise from different node states (e.g., computing resources, residual energy) due to non-uniform working durations.
- **Resource Constraints.** Autonomous agents have limited energy, computing, and storage capacities, while also sharing constrained spectrum resources. Alongside managing their own motion control, they must execute tasks, further straining their available resources.

Given the transformative potential of DML in enhancing the capabilities of AASs and considering their unique characteristics, it is essential to conduct a comprehensive review to identify these characteristics, summarize recent research advancements, and outline potential use cases. To address this need, this survey examines the features of typical AASs from both communication and control perspectives, including UAVs, UGVs, USVs, and UUVs. Then, we provide an overview of the basic ML models used in AASs, examine their limitations for efficient application, and highlight the role of DML in overcoming these challenges. Furthermore, this survey further explores use cases and key techniques for DML in real-world scenarios, particularly focusing on two representative DML techniques: FL and MARL. This work aims to provide valuable insights for researchers and practitioners engaged in this field. The contributions of this survey can be summarized as follows:

- We introduce four types of representative AASs, analyzed through a unified framework covering their basic characteristics, communication models, and dynamic models. This unified approach facilitates a better understanding of the diverse nature of AASs and their operational principles.
- We review the basic ML models used in AASs, examine why existing basic ML models are not sufficient for efficient application in AASs, and outline the essential requirements and metrics necessary for the successful deployment of DML in AASs.
- We present recent advancements in FL and MARL as applied to AASs, including use cases and key techniques, thereby providing a thorough overview of current technological progress and identifying gaps in the literature for future exploration.

The remainder of this paper is organized as follows. Section II reviews related surveys on AASs and DML, highlighting the distinctions between this survey and previous works. Section III presents a unified perspective on four representative types of AASs, covering their fundamental characteristics, communication models, and dynamic models. Section IV examines basic ML models designed for AASs and discusses the requirements for deploying DML in these systems. Sections V and VI respectively explore recent advancements in FL and MARL for AASs, including

key techniques and use cases. Section VII outlines promising research directions, while Section VIII concludes the paper.

II. RELATED SURVEYS

There have already been numerous reviews on AASs, which can generally be categorized into communication and networking [1], [2], [3], [4], [5], [6], [26], [27], [34], [38], localization and navigation [7], [8], [9], [28], [29], [30], [31], perception [10], [11], [12], [13], [14], [32], [33], [34], control and path planning [15], [16], [17], [18], [19], [35], [36], [37], [39], [40], and task assignment [20], [21], [22], [23], [24], [25], [41], [42], [43]. Among these, several surveys have specifically focused on the application of ML in AASs and discussed recent advancements in this emerging area, such as ML-enabled communication and networking [26], [27], [34], [38], ML-enabled localization and navigation [28], [29], [30], [31], ML-enabled perception [32], [33], [34], ML-enabled control and path planning [35], [36], [37], [39], [40], and ML-enabled task assignment [41], [42], [43]. We summarize these surveys on AASs in Table I. To enhance prediction quality and address complex applications, ML has transitioned from a centralized paradigm to a distributed one, known as DML. While there are numerous surveys on DML, they primarily concentrate on wireless networks, IoT, and so on. Surveys specifically covering DML for AASs remain relatively limited. In Table II, we summarize the characteristics and differences between existing surveys and the present survey.

1) *DML for Wireless Networks:* DML is critical in wireless communications, addressing challenges of large-scale, geographically dispersed, and highly dynamic networks. The survey in [44] explores recent DML applications in power control, spectrum management, and edge-cloud computing, analyzing factors like optimality, accuracy, and communication overhead. In [45], the authors emphasize the importance of continuous data and model updates for scalable ML inference under dynamic conditions, and propose optimizing communication efficiency. The work in [46] integrates FL with meta learning to improve model adaptability without compromising privacy, presenting frameworks for communication-efficient learning. The survey in [47] uniquely combines cyber-physical-social systems with FL, highlighting privacy-preserving mechanisms for handling heterogeneous data. The authors in [48] and [49] provide comprehensive reviews on MARL and DRL, focusing on their applications in emerging wireless technologies like UAV networks, edge computing, and network security.

2) *DML for IoT and Edge Computing:* The proliferation of IoT and distributed computing has driven the development of DML to enhance IoT services and applications. The authors in [50] survey DML's role in critical IoT services like data sharing, computation offloading, and mobile crowdsensing, as well as in applications like smart healthcare and autonomous vehicles. In [51], the authors review end-edge-cloud computing (EECC) enabled DML, categorizing optimization techniques that enhance distributed training and inference. The work in [52] explores FL in edge environments

TABLE I
THE DESCRIPTION OF RELATED SURVEYS ON AASS

Work	Category	Object	Main Focus	ML	DML
[1]	Communication and Networking	UAV	Swarm networking, including channel access, routing, and jamming	×	×
[2]	Communication and Networking	UAV	Maritime communication	×	×
[3]	Communication and Networking	UGV	Named data networking for dynamic topology	×	×
[4]	Communication and Networking	USV	Maritime networking, including sensing, communication, and computing	×	×
[5]	Communication and Networking	UUV	Underwater acoustic communication	×	×
[6]	Communication and Networking	UUV	Underwater optical communication	×	×
[26]	Communication and Networking	UAV	RL for networking, including data collection, resource allocation	✓	×
[27]	Communication and Networking	UGV	ML-aided cognitive radio	✓	×
[34]	Communication and Networking	UAV	ML-aided communication	✓	×
[38]	Communication and Networking	UGV	ML-aided network security	✓	×
[7]	Localization and Navigation	UAV	Localization for swarm, in terms of data, architecture, and algorithms	×	×
[8]	Localization and Navigation	UUV	Underwater navigation, including state estimation, and SLAM	×	×
[9]	Localization and Navigation	UUV	Cost-effective localization strategies	×	×
[28]	Localization and Navigation	UGV	Classical and ML methods for navigation	✓	×
[29]	Localization and Navigation	UAV	Optimization-based and ML-based algorithms for navigation	✓	×
[30]	Localization and Navigation	UUV	DL for inertial navigation	✓	×
[31]	Localization and Navigation	USV	RL and IL for navigation	✓	×
[10]	Perception	UAV	Integration of wireless sensor networks, UAV swarms, and crowdsensing	×	×
[11]	Perception	UAV	Data collection, target recognition, and tracking in complex areas	×	×
[12]	Perception	UGV	Cooperative perception via multi-sensor fusion methods	×	×
[13]	Perception	UAV/UGV	Cross-domain collaboration for data collection	×	×
[14]	Perception	USV/UUV	Marine data collection	×	×
[32]	Perception	UGV	DL for processing image and point cloud data	✓	×
[33]	Perception	USV	Object detection for maritime surveillance and navigation	✓	×
[34]	Perception	UAV	ML for automation and precision, particularly in sensing	✓	×
[15]	Control and Path Planning	UAV	Swarm formation control	×	×
[16]	Control and Path Planning	UAV	Swarm intelligence for path planning	×	×
[17]	Control and Path Planning	USV	Path planning, considering sea conditions like wind, waves, and currents	×	×
[18]	Control and Path Planning	UUV	Dynamic characteristics and trajectory tracking control methods	×	×
[19]	Control and Path Planning	UUV	Formation transformations in varied underwater tasks.	×	×
[35]	Control and Path Planning	UAV	ML for path planning	✓	×
[36]	Control and Path Planning	UUV	ML for path planning, emphasizing practical deployment challenges	✓	×
[37]	Control and Path Planning	UUV	ML-based control, emphasizing trajectory tracking and formation control	✓	×
[39]	Control and Path Planning	UGV	RL for motion planning	✓	×
[40]	Control and Path Planning	USV	RL for path planning	✓	×
[20]	Task Assignment	AAS	Market-, behavior-, and optimization-based task assignment	×	×
[21]	Task Assignment	AAS	Multi-tasking scenarios task assignment	×	×
[22]	Task Assignment	UAV	Task assignment and optimization algorithms	×	×
[23]	Task Assignment	UAV	Dynamic task assignment approaches for ground targets	×	×
[24]	Task Assignment	UUV	Task assignment in uncertain and unreliable underwater environments	×	×
[25]	Task Assignment	UUV	Centralized and distributed task assignment algorithms for complex tasks	×	×
[41]	Task Assignment	UAV	ML-aided task assignment	✓	×
[42]	Task Assignment	UAV	ML for cooperative tasking model for heterogeneous swarms	✓	×
[43]	Task Assignment	AAS	RL for dynamic task assignment	✓	×

for 6G, addressing challenges and technical strategies. The authors in [53] discuss pervasive artificial intelligence (AI), integrating DL and RL to improve distributed computations and communication efficiency across IoT, edge, and cloud systems.

3) *DML for AAS*: Several pioneering works have explored the application of DML in AASs. Compared to previous surveys on DML for wireless networks, DML for AAS exhibits notable distinctions, including agent dynamics, uncertain communication links, and challenging operational environments. The work in [58] categorizes UAV swarm tasks into coverage, adversarial search and gaming, computational offloading, communication, and target-driven navigation, highlighting the role of MARL in these applications. Similarly, [54] introduces four DML frameworks and explores their applications in communication, computing, caching, sensing, and trajectory design for UAV swarms. In [56], advancements in FL for UGV swarms are reviewed, covering centralized and decentralized frameworks, privacy techniques, and specific

applications, while the work in [57] examines the role of 5G in enhancing DML for autonomous vehicles, emphasizing vehicle-to-vehicle (V2V) and vehicle-to-everything (V2X) communication to improve network perception, safety, and privacy. The authors in [55] focus on MARL in AASs, categorizing the literature by application scenarios and identifying future research directions. However, existing surveys lack a unified perspective to systematically identify the commonalities and differences among various types of AASs, as they primarily focus on individual categories. This fragmentation impedes researchers from effectively transferring successful methodologies between different AAS systems. Moreover, the two most promising DML paradigms, FL and MARL, have not been comprehensively reviewed in the context of their core technologies and applications for AASs. To bridge these critical gaps, this survey provides a holistic analysis, aiming to facilitate cross-system knowledge transfer and offer valuable insights into the integration of FL and MARL with AASs.

TABLE II
COMPARISON OF RELATED SURVEYS ON DML

Work	Is the research object AASs?	Commonalities and differences of various AASs	FL	MARL	Brief summary
[44]	×	×	✓	✓	Techniques and applications of DML for wireless networks
[45]	×	×	✓	×	Communication-efficient DML algorithms for wireless networks
[48]	×	×	×	✓	Applications of MARL for future Internet
[46]	×	×	✓	×	Applications of FL and meta learning for wireless networks
[47]	×	×	✓	×	FL in cyber-physical-social systems to enhance data privacy and efficiency
[52]	×	×	✓	×	Combining FL and edge computing to advance ubiquitous intelligence
[50]	×	×	✓	✓	Applications of DML algorithms for IoT
[51]	×	×	✓	×	Key techniques and applications of DML for EECC
[54]	Partially (UAV)	×	✓	✓	Applications of DML algorithms for UAV swarms
[55]	✓	×	×	✓	Applications of MARL for AAS
[56]	Partially (UGV)	×	×	✓	Key techniques and applications of FL for UGV Swarms
[57]	Partially (UGV)	×	×	✓	ML for vehicular communication, focusing on data sharing
[49]	×	×	×	✓	Application of MARL in wireless networks
[53]	×	×	✓	✓	Applications of DML algorithms for IoT
[58]	Partially (UAV)	×	×	✓	Application of MARL in UAV swarm
This survey	✓	✓	✓	✓	Key techniques and applications of DML for AAS

III. COMMONALITIES AND DISTINCTIONS AMONG VARIOUS AASS

As previously discussed, existing studies on DML—including FL and MARL—for AASs predominantly concentrate on methods developed for a specific type of system. This focus has resulted in a fragmented research landscape, limiting the effective transfer and generalization of successful methodologies across heterogeneous AASs. While many FL and MARL techniques exhibit inherent potential for cross-domain adaptation, their direct application to different AAS types is often constrained by fundamental system-level disparities. In practice, beyond payload differences, two primary factors significantly contribute to this heterogeneity. First, there are substantial variations in dynamic models. For instance, UAVs operate in three-dimensional space with high maneuverability and minimal environmental contact, whereas UGVs are restricted to two-dimensional ground movement, subject to surface interactions and friction. Second, communication modalities vary considerably: UAVs typically employ high-speed, low-latency radio links, whereas UUVs rely on narrowband, high-latency acoustic communication due to the constraints of underwater environments. These divergences highlight the need for rigorous analysis and systematic characterization of both the shared features and the system-specific attributes of different AAS types. Such analysis serves as a critical foundation for identifying transferable components of FL and MARL algorithms and for determining the necessary adaptations required to accommodate specific system characteristics. Ultimately, this enables the robust and efficient development and deployment of DML methodologies across heterogeneous AAS environments.

In this section, we aim to provide a unified perspective on the commonalities and distinctions among various AASs, focusing on four representative categories: UAV, UGV, USV, and UUV swarms. We begin by presenting the typical agent types within each AAS, highlighting their motion characteristics and commonly integrated sensors. Subsequently, we propose a generalized model of inter-agent communication and control, followed by an in-depth analysis of both their structural differences and system-specific traits.

A. Brief Review of Various AASs

1) *UAV Swarm*: UAVs primarily fall into two categories: fixed-wing and rotor-wing, each with distinct characteristics and operational speed ranges.

- *Fixed-wing UAVs*. These UAVs resemble traditional airplanes with fixed wings and are usually designed for long-distance flights and high-speed operations. They typically have a speed range of 40 to 300 km/h.
- *Rotor-wing UAVs*. These UAVs use rotors for lift and maneuverability. They are often used for aerial photography, surveillance, and short-distance missions. Their speed generally ranges from 20 to 60 km/h.

Typically, UAVs are equipped with various sensors, such as global positioning systems (GPS), inertial measurement units (IMUs), cameras (RGB, infrared, thermal), light detection and ranging (LiDAR), ultrasonic sensors, barometric pressure sensors, and magnetometers. Table III summarizes the commonly used sensors in AASs.

2) *UGV Swarm*: UGVs are mainly categorized into wheeled and tracked types, determined by their locomotion mechanisms.

- *Wheeled UGVs*. These UGVs use wheels for movement, providing high speed and efficient operation on flat and relatively smooth surfaces. Their speeds typically range from 10 to 120 km/h.
- *Tracked UGVs*. Equipped with tracks, these UGVs are designed for rugged and uneven terrains, offering better traction and stability. Their speeds generally range from 5 to 60 km/h.

UGVs are usually equipped with GPS, IMUs, cameras (RGB, stereo), LiDAR, radar, ultrasonic sensors, proximity sensors, and various environmental sensors.

3) *USV Swarm*: USVs are mainly classified into planing hull and displacement hull types, distinguished by their hull design and propulsion features.

- *Planing hull USVs*. These USVs are designed to rise and glide on the water surface at high speeds. They are typically used for fast-response missions and can achieve speeds ranging from 20 to 60 km/h.

TABLE III
SENSORS USED IN AASS AND THEIR FUNCTIONALITIES

Sensor	Role	UAV	UGV	USV	UUV
GPS	Provides precise positioning and navigation.	✓	✓	✓	✓
IMU	Stabilizes and controls the movement of the vehicle.	✓	✓	✓	✓
Cameras	Used for visual monitoring, target identification, and environmental surveillance	✓	✓	✓	✓
LiDAR	Used for obstacle detection, terrain mapping, and 3D modeling	✓	✓		
Ultrasonic Sensors	Utilized for close-range obstacle detection and altitude maintenance	✓	✓		
Barometric Pressure Sensors	Used for altitude measurement	✓			
Magnetometers	Employed for heading and orientation detection	✓			
Radar	Used for long-range object detection and tracking		✓	✓	
Proximity Sensors	Detect nearby objects to avoid collisions		✓		
Environmental Sensors	Monitor surroundings such as temperature, humidity, and pressure		✓	✓	✓
Sonar	Used for underwater mapping, obstacle detection, and terrain following			✓	✓
AIS	Tracks other vessels to avoid collisions			✓	
INS (IMU + DVL)	Provides accurate underwater navigation				✓
Depth Sensors	Measure the depth				✓
Chemical Sensors	Detect various water quality parameters like dissolved oxygen and pH levels				✓

- *Displacement hull USVs.* These USVs have hulls that displace water to move, similar to traditional boats and ships. They are designed for stability and efficiency over longer durations, often used for survey and monitoring tasks. Their speeds typically range from 5 to 20 km/h.

USVs are generally equipped with GPS, IMUs, radar, sonar (echo sounder), cameras (RGB, thermal), environmental sensors, and automatic identification system (AIS).

4) *UUV Swarm:* UUVs are primarily classified into two categories based on their propulsion mechanisms: propeller-driven and bio-inspired UUVs.

- *Propeller-driven UUVs.* These UUVs use traditional propellers for propulsion and are designed for efficient and controlled underwater navigation. Their speeds typically range from 5 to 10 km/h. Some military UUVs, such as Russia's Poseidon, can reach speeds of several hundred km/h.
- *Bio-inspired UUVs.* Mimicking the movement of aquatic animals, these UUVs use fins, undulating bodies, or other bio-inspired mechanisms for propulsion. They generally have slower speeds ranging from 1 to 5 km/h but offer superior maneuverability and efficiency in complex underwater environments.

UUVs are commonly equipped with sensors such as sonar (side-scan, multi-beam, forward-looking), inertial navigation systems (INS) combining IMUs and Doppler velocity logs (DVLs), depth sensors, cameras (low-light, high-definition), and chemical sensors.

B. Commonalities and Distinctions Among Various AASS in Communications

1) *Generalized Communication Model:* We consider a generalized communication model based on the classical Shannon's Theorem that defines the basic limits of communication in various environmental conditions, which is given by

$$C = B \log_2 \left(1 + \frac{PH}{N} \right), \quad (1)$$

where C represents the channel capacity, B is the bandwidth, P is the signal power, N denotes the noise power, and H

represents the channel gain. This model shows the fundamental communication capabilities between autonomous agents. However, although the aforementioned unified communication model can serve as a foundational framework for all AASSs, notable differences in their communication characteristics emerge in practical applications. These differences are primarily manifested in the following aspects:

a) *Medium, frequency bands, bandwidth, and transmission rates:*

- UAVs primarily employ electromagnetic waves for communication, operating in frequency bands such as 2.4 GHz, 5.8 GHz, L-band, and S-band. Bandwidth ranges from a few MHz to tens of MHz, enabling transmission rates from a few Mbps to tens of Mbps.
- UGVs also utilize electromagnetic waves for communication, typically in 2.4 GHz, 5.9 GHz, and ultra high frequency (UHF) bands. Bandwidth ranges from a few MHz to tens of MHz, with transmission rates between a few Mbps and tens of Mbps.
- USVs rely on electromagnetic waves, predominantly in the VHF band, with bandwidths of 12.5 kHz or 25 kHz and transmission rates in the range of a few kbps.¹
- UUVs usually adopt acoustic waves for communication due to the constraints of underwater electromagnetic propagation. Frequency bands range from a few kHz to hundreds of kHz, with bandwidths between a few kHz and tens of kHz, supporting transmission rates from a few bps to tens of kbps.

b) *Channel characteristics:*

- Communications among UAVs primarily occur in line-of-sight (LOS) channels. However, signal performance may deteriorate in urban environments or under adverse atmospheric conditions due to obstructions such as buildings or weather-related phenomena, which disrupt LOS propagation.
- Communications among UGVs are significantly affected by ground obstacles, including buildings, walls, and uneven road surfaces. These challenges can cause signal

¹Given the extensive operations of ships in international waters and their unique requirements, international organizations have established strict regulations for the frequency bands allocated to maritime communication [59].

attenuation, reflection, or multipath effects, particularly in urban or complex terrains. Additionally, spectrum congestion and wireless interference in urban areas further restrict the reliability and range of the communications among UGVs.

- Communications among USVs are significantly influenced by the marine environment. While electromagnetic waves can propagate effectively over long distances across the water surface, signal stability is often compromised by surface wave fluctuations, the motion of USVs, and interference from other marine devices. Adverse weather conditions, such as strong winds or heavy seas, further degrade communication quality and reliability.
- Communications among UUVs rely on underwater acoustic wave propagation, which is influenced by water temperature, salinity, depth, ocean currents, and turbulence. Furthermore, underwater noise from marine life, ship engines, and other sources degrades the quality of communications and increases bit error rates.

Hence, we will subsequently present the communication models for intra-cluster communications within different AASs, respectively. In fact, the differences in communication within AASs largely stem from the characteristics of their communication channels, particularly the signal fading properties. These can be categorized into large-scale fading (primarily path loss) and small-scale fading, both of which collectively influence the value of H in Eq. (1). Since large-scale fading, particularly path loss, typically plays a more dominant role in long-distance or less complex propagation environments, and given the space limitations of this paper, this section focuses mainly on the discussion of path loss in communications among autonomous agents within different AASs.

2) *Communication Model of UAV Swarm*: Due to the operating altitude of UAV swarms, there is often no obstruction between UAVs. Therefore, the links between UAVs are considered to be LOS channels [60]. The path loss of the UAV-UAV link can be calculated as

$$p_{\text{pl}}^{\text{u}} = 20 \log \left(\frac{4\pi f_c d_{\text{u}}}{c} \right) + G_{\text{u}}, \quad (2)$$

where f_c represents the carrier frequency, d_{u} denotes the distance between two UAVs, c is the speed of light in vacuum, and G_{u} represents additional attenuation factors.

3) *Communication Model of UGV Swarm*: According to [61], in rural and highway scenarios, the path loss between UGVs can be calculated as

$$p_{\text{pl}}^{\text{g}} = G_{\text{g1}} + \Delta G(\sigma) + 20 \log \left(\frac{\lambda}{4\pi} \left| \frac{\exp(-jk_0 d_{-}^{\text{g}})}{d_{-}^{\text{g}}} + \zeta \frac{\exp(-jk_0 d_{+}^{\text{g}})}{d_{+}^{\text{g}}} \right| \right), \quad (3)$$

where λ represents the wavelength at the center frequency, and $k_{\text{u}} = 2\pi/\lambda$. d_{-}^{g} and d_{+}^{g} are the direct and reflected path distances determined by the lateral distance and from the antenna height h . ζ represents the effective ground reflection coefficient, and $\Delta G(\sigma)$ is zero-mean white Gaussian noise

(AWGN) with standard deviation σ . In urban scenarios, the path loss of UGVs follows

$$p_{\text{pl}}^{\text{g}} = G_{\text{g2}} + n10 \log \left(\frac{d_{\text{g}}^{\text{g}}}{d_0^{\text{g}}} \right) + \Delta G(s), \quad (4)$$

where n is a path gain exponent, while $\Delta G(s)$ is zero-mean AWGN, and d_{g}^{g} is the distance between UGVs. The term G_{g2} is the path loss at a reference distance d_0^{g} in dB.

4) *Communication Model of USV Swarm*: In general, the communication links between USVs can be modeled by a three-ray path loss model [62], [63], including a direct LOS path, a reflected path from the sea surface, and a refracted path through the evaporation duct. Specifically, the path loss between two USVs can be expressed as

$$p_{\text{pl}}^{\text{s}} = \left[\frac{c}{2\pi f_c d_{\text{s}}} (1 + \varrho) \right]^{-2}, \quad (5)$$

with

$$\varrho = 2 \sin \left(\frac{2\pi h_{\text{T}}^{\text{s}} h_{\text{R}}^{\text{s}} f_c}{c d_{\text{s}}} \right) \sin \left(\frac{2\pi (h' - h_{\text{T}}^{\text{s}})(h' - h_{\text{R}}^{\text{s}}) f_c}{c d_{\text{s}}} \right), \quad (6)$$

where d_{s} is the distance between two USVs. h' , h_{T}^{s} , and h_{R}^{s} denote the effective evaporation duct height, transmit antenna height, and receive antenna height, respectively.

5) *Communication Model of UUV Swarm*: Unlike the previously mentioned AASs, which rely on radio-based communication, acoustic communication is widely used in UUV swarms due to the significant attenuation of radio waves in underwater environments [64]. Let $A(d_{\text{a}}, f_{\text{a}})$ be the acoustic path loss as a function of communication frequency f_{a} and distance d_{a} between two UUVs, which is given by

$$A(d_{\text{a}}, f_{\text{a}}) = d_{\text{a}}^k a(f_{\text{a}})^{d_{\text{a}}}, \quad (7)$$

where k is the spreading factor, while $a(f_{\text{a}})$ represents the absorption coefficient given by

$$10 \log a(f_{\text{a}}) = \frac{0.11 f_{\text{a}}^2}{1 + f_{\text{a}}^2} + \frac{44 f_{\text{a}}^2}{4100 + f_{\text{a}}^2} + 2.75 \cdot 10^{-4} f_{\text{a}}^2 + 0.003. \quad (8)$$

C. Commonalities and Distinctions Among Various AASs in Control

1) *Generalized Motion and Control Model*: If we consider the 6 degrees of freedom (DOF) rigid-body motion equations or a state-space representation in control theory, treating environmental disturbances, constraints, and actuator effects as generalized “inputs” and “disturbances,” then each autonomous agents, including UAVs, UGVs, USVs, and UUVs, can all be “formalized” in a highly abstract, unified control framework. We can derive analogous rigid-body dynamics from the fundamental Newton–Euler equations

$$\begin{cases} \dot{\mathbf{p}} = \mathbf{v}, \\ \dot{\mathbf{v}} = \mathbf{f}(\mathbf{v}, \boldsymbol{\omega}, \mathbf{u}, t), \\ \dot{\boldsymbol{\Theta}} = \mathbf{R}(\boldsymbol{\Theta}) \boldsymbol{\omega}, \\ \dot{\boldsymbol{\omega}} = \mathbf{g}(\mathbf{v}, \boldsymbol{\omega}, \mathbf{u}, t), \end{cases} \quad (9)$$

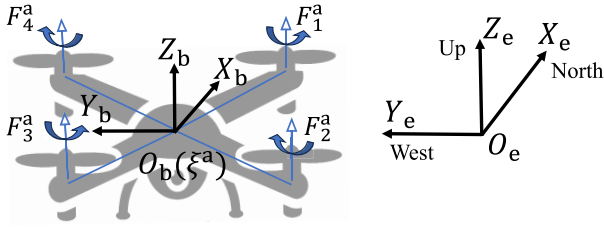


Fig. 2. The Dynamic Model of UAV [65].

where $\mathbf{p}$ represents position, Θ denotes orientation (e.g., Euler angles or quaternions), $\mathbf{v}$ is the linear velocity, $\boldsymbol{\omega}$ the angular velocity, $\mathbf{u}$ the control input, and $\mathbf{f}$, $\mathbf{g}$ capture the system dynamics. By appropriately modeling external forces (e.g., aerodynamic/hydrodynamic forces, gravity, buoyancy, rolling resistance), environmental disturbances (e.g., wind fields, ocean currents), and actuator characteristics (e.g., propellers, wheels, motors, rudders), we can arrive at a unified set of differential equations. By leveraging 6-DOF equations and general state-space frameworks, one can represent different autonomous agents in a “unified” mathematical form. However, in actual engineering practice and for precise control, each type of autonomous agent exhibits significant differences in the following aspects:

a) Operating medium and fluid dynamics:

- UAVs deal with aerodynamic lift, drag, and wind disturbances in air.
- USVs/UUVs cope with buoyancy, hydrodynamic drag, and ocean currents.
- UGVs face rolling resistance, wheel-ground friction.

b) Types of control inputs and actuator mechanisms:

- UAVs manipulate thrust and attitude via propellers or control surfaces (e.g., ailerons, rudder).
- UGVs steer and accelerate via wheel torque and steering angles.
- USVs/UUVs use propellers, thrusters, rudders, or other hydrodynamic devices.

c) Kinematics and nonholonomic constraints:

- Certain UGVs cannot move laterally (nonholonomic constraints).
- UAVs have specific attitude-velocity coupling (especially multirotors).
- UUVs involve buoyancy control, multiple degrees of freedom, and underwater constraints.

Despite the similar form of the equations, each domain demands customized modeling, parameter identification, and control strategies. Large disparities in environment, dynamics, actuators, noise, nonholonomic constraints, and safety requirements lead to significant variation in practical modeling and controller design. Thus, real-world systems require substantial tailoring for each autonomous agent. In the following, we present the corresponding dynamic and control models of different types of autonomous agents, respectively.

2) Motion and Control Model of UAVs: In the following, we use the more widely applied quadrotor as an example to illustrate UAV’s dynamic features.

The quadrotor is a typical underactuated system with 6 DOF. These include translational motion along the three axes of a spatial rectangular coordinate system and rotational motion around these axes. However, it has only four controllable inputs, i.e., the speeds of the four motors. Fig. 2 illustrates the body coordinate system and the geographic coordinate system of the UAV. The Euclidean position and Euler angles of the UAV in the geographic frame are represented by $\boldsymbol{\xi}^a = [x^a, y^a, z^a]^T$ and $\boldsymbol{\eta}^a = [\phi^a, \theta^a, \psi^a]^T$, respectively. Then the dynamic model can be given by [65]

$$\begin{cases} m^a \ddot{x}^a = (\cos \phi^a \sin \theta^a \cos \psi^a + \sin \phi^a \sin \psi^a) u_t^a - K_1 \dot{x}^a \\ m^a \ddot{y}^a = (\cos \phi^a \sin \theta^a \sin \psi^a - \sin \phi^a \cos \psi^a) u_t^a - K_2 \dot{y}^a \\ m^a \ddot{z}^a = (\cos \phi^a \cos \theta^a) u_t^a - K_3 \dot{z}^a - m^a g \\ J_1 \ddot{\phi}^a = d_1^a + l \tau_1^a - K_4 l \dot{\phi}^a \\ J_2 \ddot{\theta}^a = d_2^a + l \tau_2^a - K_5 l \dot{\theta}^a \\ J_3 \ddot{\psi}^a = d_3^a + c \tau_3^a - K_6 \dot{\psi}^a, \end{cases} \quad (10)$$

where m^a is the mass, and $K_i, i = \{1, \dots, 6\}$ represent the aerodynamic damping coefficient. g denotes the acceleration due to gravity, and $J_i, i = \{1, 2, 3\}$ represent the moments of inertia. $l \in \mathbb{R}^+$ and $d_i^a, i = \{1, 2, 3\}$ represent the distance between the epicenter of the UAV and the rotor axis, and the unknown time-varying external disturbances, respectively. u_t^a is the total thrust, and τ_1^a, τ_2^a , and τ_3^a are the three rotational forces produced by the four rotors. Additionally, $c \in \mathbb{R}^+$ denotes a force-to-moment factor. Let $F_i^a, i = \{1, 2, \dots, 4\}$ represent the thrusts generated by each motor. The control input can be defined as $\mathbf{C}^a = [u_t^a, \tau_1^a, \tau_2^a, \tau_3^a]^T$, whose relationship with F_i^a is demonstrated as

$$\begin{cases} u_t^a = F_1^a + F_2^a + F_3^a + F_4^a \\ \tau_1^a = -F_1^a - F_2^a + F_3^a + F_4^a \\ \tau_2^a = -F_1^a + F_2^a + F_3^a - F_4^a \\ \tau_3^a = -F_1^a + F_2^a - F_3^a + F_4^a. \end{cases} \quad (11)$$

3) Motion and Control Model of UGVs: Let $\boldsymbol{\xi}^g = [x^g, y^g, \phi^g]^T$ denote the state of the UGV, where (x^g, y^g) represents the position, and ϕ^g denotes the heading angle. As depicted in Fig. 3, the dynamic model of the UGV can be given by [66]

$$\begin{cases} \dot{x}^g = v^g \cos \phi^g \\ \dot{y}^g = v^g \sin \phi^g \\ \dot{\phi}^g = \omega = \dot{\beta} + r \\ m^g \dot{v}_x^g = F_{xf}^g \cos \delta^g - F_{yf}^g \sin \delta^g + F_{xr}^g - F_x^r + m^g v_y^g r \\ m^g \dot{v}_y^g = F_{xf}^g \sin \delta^g + F_{yf}^g \cos \delta^g + F_{yr}^g - F_y^d - m^g v_x^g r \\ J_z \dot{r} = l_f F_{xf}^g \sin \delta^g + l_f F_{yf}^g \cos \delta^g - l_r F_{yr}^g + \Delta M_z - M_z^r, \end{cases} \quad (12)$$

v^g is the speed of the UGV, consisting of the longitudinal velocity v_x^g and lateral velocity v_y^g . ω represents the angular velocity, while r is the yaw rate. β is the side slip angle of the mass center, while m^g is the mass of the UGV. The term J_z denotes the moment of inertia about the vertical axis, and δ^g is the front steering angle. The distances between the mass center and the front and rear axles are denoted by l_f and l_r , respectively. The term F_x^r denotes a longitudinal resistance, while F_y^d and M_z^r denote lateral and heading disturbances, respectively. Define $(F_{xf}^g, F_{xrf}^g, F_{xlr}^g, F_{xrr}^g)$ and $(F_{yf}^g, F_{yrf}^g, F_{ylr}^g, F_{yrr}^g)$

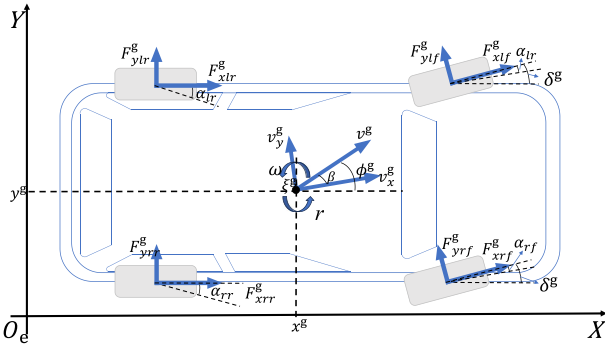


Fig. 3. The Dynamic Model of UGV [66].

as the sets of longitudinal and lateral forces from each wheel, respectively. Hence, the longitudinal forces of the front and rear wheels F_{xf}^g and F_{xr}^g , and the lateral force of the front and rear wheels F_{yf}^g and F_{yr}^g , as well as the active yaw moment ΔM_z are calculated as

$$\begin{cases} F_{xf}^g = F_{xlf}^g + F_{xrf}^g \\ F_{xr}^g = F_{xlr}^g + F_{xrr}^g \\ F_{yf}^g = F_{yfl}^g + F_{yfr}^g \\ F_{yr}^g = F_{yrl}^g + F_{yrr}^g \\ \Delta M_z = \frac{W}{2} [F_{xrf}^g - F_{xlr}^g - F_{xlr}^g + F_{xrr}^g] \end{cases} \quad (13)$$

where W denotes the track width of the UGV. Let $i = \{fl, fr, rl, rr\}$ be the index of the wheel, and the torque balance equation determining the rotational dynamics of the four wheels is given by

$$J_\omega \dot{\omega}_i = T_{di} - T_{bi} + T_{fi} - r_\omega F_{xi}^g, \quad (14)$$

where J_ω denotes the moment of the wheel inertia, while $\dot{\omega}_i$ is the rotational angular acceleration. T_{di} and T_{bi} represent the driving and braking torques on a tire, respectively, while T_{fi} denotes the rolling resisting moment. r_ω is the tire radius, while F_{xi}^g denotes the longitudinal force of a tire. According to the Dugoff tire model, the longitudinal force F_{xi}^g and the lateral force F_{yi}^g are given by

$$\begin{cases} F_{xi}^g = \frac{C_\sigma \sigma_i}{1 + \sigma_i} f(\lambda_i), \\ F_{yi}^g = \frac{C_\alpha \tan \alpha_i}{1 + \sigma_i} f(\lambda_i), \end{cases} \quad (15)$$

where σ_i and α_i are the wheel slip ratio and slip angle, respectively. C_σ and C_α represent the longitudinal and lateral tire stiffness, respectively, while F_{zi}^g denotes the vertical load force of a tire. μ is the ground adhesion coefficient, while $\lambda_i = \frac{\mu F_{zi}^g (1 + \sigma_i)}{2\sqrt{(C_\sigma \sigma_i)^2 + (C_\alpha \tan \alpha_i)^2}}$ represents the attachment reserve coefficient. And $f(\lambda_i)$ is defined as

$$f(\lambda_i) = \begin{cases} \lambda_i(2 - \lambda_i), & \lambda_i < 1 \\ 1, & \lambda_i \geq 1 \end{cases}$$

σ_i and α_i are represented as

$$\sigma_i = \begin{cases} 1 - \frac{v_x^g}{r_\omega \omega_i}, & v_x^g < r_\omega \omega_i \\ \frac{r_\omega \omega_i}{v_x^g} - 1, & v_x^g > r_\omega \omega_i \end{cases} \quad (16)$$

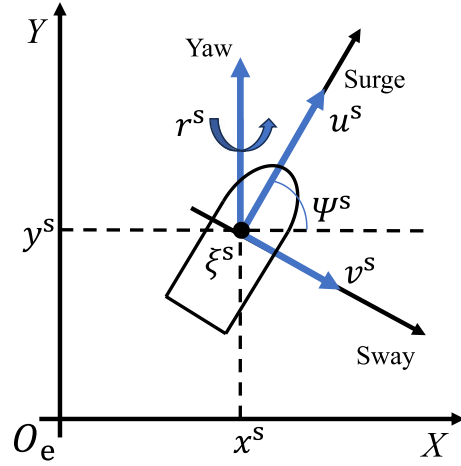


Fig. 4. The Dynamic Model of USV [67].

and

$$\begin{cases} \alpha_f = \beta + r l_f / v_x^g - \delta^g, \\ \alpha_r = \beta - r l_r / v_x^g, \end{cases} \quad (17)$$

respectively. The control input is designed as $C^g = [v_x^g, \omega]^T$.

4) *Motion and Control Model of USVs*: Fig. 4 shows the schematic depiction of a USV [67]. Let $\xi^s = [x^s, y^s, \psi^s]^T$ and $v^s = [u^s, v^s, r^s]^T$ represent the position and velocity vectors, respectively, where (x^s, y^s) represents the USV's position in the earth-fixed inertial coordinate system, and ψ^s is the heading angle, while u^s , v^s , and r^s are the surge, sway, and yaw motions of the USV in the body-fixed coordinate system. Therefore, the dynamic model of the USV can be given by

$$\begin{cases} \dot{\xi}^s = R(\psi^s) v^s, \\ M^s \dot{v}^s + Co(v^s) v^s + D v^s = C^s + e^s, \end{cases} \quad (18)$$

where $R(\psi^s)$ is defined as

$$R(\psi^s) = \begin{bmatrix} \cos \psi^s & -\sin \psi^s & 0 \\ \sin \psi^s & \cos \psi^s & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (19)$$

M^s is the mass matrix of the USV, including the rigid mass and added mass. Each element of this matrix is denoted by m_{ij}^s . $Co(v^s)$ denotes the Coriolis and centripetal terms:

$$Co(v^s) = \begin{bmatrix} 0 & 0 & c_{13} \\ 0 & 0 & c_{23} \\ c_{31} & c_{32} & 0 \end{bmatrix},$$

where $c_{13} = -m_{22}^s v^s - \frac{1}{2}(m_{23}^s + m_{32}^s) r^s$, $c_{23} = m_{11}^s u^s$, $c_{31} = -c_{13}$, and $c_{32} = -c_{23}$. D is the nonlinear damping matrix, which is given by

$$D = \begin{bmatrix} d_{11}^s & 0 & 0 \\ 0 & d_{22}^s & d_{23}^s \\ 0 & d_{32}^s & d_{33}^s \end{bmatrix}, \quad (20)$$

where d_{11}^s , d_{22}^s , d_{33}^s , d_{23}^s , and d_{32}^s are hydrodynamic derivatives, including both linear and nonlinear damping terms. The term $e^s = [e_u^s, e_v^s, e_r^s]^T$ represents disturbances caused by wind, ocean currents, and waves, where e_u^s , e_v^s , and e_r^s are the disturbances in surge, sway, and yaw motions, respectively.

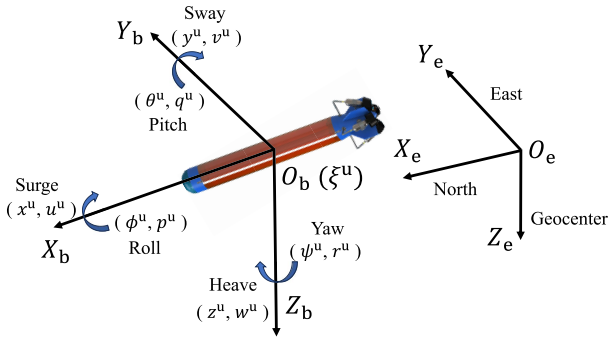


Fig. 5. The Dynamic Model of UUV.

The control input vector is $\mathbf{C}^s = [\tau_u^s, 0, \tau_r^s]^T$, where τ_u^s is the surge force and τ_r^s is the yaw moment.

5) *Motion and Control Model of UUVs*: Fig. 5 shows an underactuated UUV in reference coordinates. Let $\xi^u = [x^u, y^u, z^u, \theta^u, \psi^u]^T$ denote the position vector, where (x^u, y^u, z^u) represents the position, while θ^u and ψ^u represent the attitudes. $\mathbf{v}^u = [u^u, v^u, w^u, q^u, r^u]^T$ represents the velocity vector, including the linear velocities u^u, v^u, w^u , and angular velocities q^u, r^u in the body-fixed frame. In practice, the roll motion is usually ignored due to the structure being symmetrical about its main symmetric axis. According to [37], the dynamic model can be given by

$$\begin{cases} \dot{\xi}^u = \mathbf{R}(\xi^u) \mathbf{v}^u \\ \mathbf{M}^u \dot{\mathbf{v}}^u + \mathbf{Co}(\mathbf{v}^u) \mathbf{v}^u + \mathbf{D}(\mathbf{v}^u) \mathbf{v}^u + \mathbf{g}(\xi^u) = \mathbf{C}^u + \mathbf{e}^u, \end{cases} \quad (21)$$

where $\mathbf{R}(\xi^u) = \mathbf{R}_z(\psi^u) \mathbf{R}_y(\theta^u)$ represents the rotation transformation matrix between the body-fixed frame and the Earth-fixed frame. $\mathbf{R}_z(\psi^u)$ and $\mathbf{R}_y(\theta^u)$ are given by

$$\mathbf{R}_z(\psi^u) = \begin{bmatrix} \cos \psi^u & -\sin \psi^u & 0 & 0 & 0 \\ \sin \psi^u & \cos \psi^u & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (22)$$

and

$$\mathbf{R}_y(\theta^u) = \begin{bmatrix} \cos \theta^u & 0 & \sin \theta^u & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -\sin \theta^u & 0 & \cos \theta^u & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{\cos \theta^u} \end{bmatrix}, \quad (23)$$

respectively. Similar to the dynamic model of the USV, $\mathbf{M}^u$ is the matrix of mass inertia and added hydrodynamic inertia, while $\mathbf{Co}(\mathbf{v}^u)$ denotes the Coriolis and centripetal matrix. $\mathbf{D}(\mathbf{v}^u)$ represents the hydrodynamic damping matrix, and $\mathbf{g}(\xi^u)$ is the restoring force and moment vector. $\mathbf{C}^u = [\tau_u^u, 0, 0, \tau_q^u, \tau_r^u]^T$ represents the control force and moment, where τ_u^u, τ_q^u and τ_r^u are the force in the surge direction and the moments in the pitch and yaw directions, respectively. $\mathbf{e}^u = [e_u^u, e_v^u, e_w^u, e_q^u, e_r^u]^T$ is the environmental disturbance force and moment, where $e_u^u, e_v^u, e_w^u, e_q^u$, and e_r^u are disturbance forces in the surge, sway, and heave directions and disturbance moments in the pitch and yaw directions, respectively.

Lessons Learned: This section presents a unified modeling framework for various AASs, covering dynamics, control, and communication. Such abstraction serves as a foundation for investigating the transferability of DML methods across heterogeneous platforms. While models such as Shannon's capacity formula and 6-DOF rigid-body dynamics offer a theoretically consistent structure, researchers must remain cautious of system-level heterogeneity masked by this formal unification. In terms of communication, although the channel capacity can be expressed uniformly as a function of bandwidth and signal-to-noise ratio, the underlying channel gain varies substantially across platforms due to differences in propagation media, transmission geometries, and onboard hardware. Regarding control modeling, despite the shared mathematical form of state-space representations, platforms differ in their degrees of freedom, actuation mechanisms, constraint formulations, and environmental disturbance characteristics—each of which critically impacts controller design and system behavior. Thus, unified modeling should be treated as an analytical foundation for capturing common structural features, rather than a substitute for system-specific understanding. Effective DML strategies must balance generality with adaptability by incorporating modular abstractions that retain fidelity to the physical and operational characteristics of each AAS type.

IV. MACHINE LEARNING MODEL DESIGN FOR AUTONOMOUS AGENT SWARM

In this section, we first review the basic use of ML for autonomous agents, and then discuss the requirements of deploying DML into AASs.

A. Basic ML Model for Autonomous Agents

Recent years, there are numerous works on using ML to enhance the capabilities of AASs, including object recognition, dynamic control, trajectory planning, navigation, communication, task allocation, information fusion, formation control, search operations, and confrontation strategies. We summarize them in Table IV.

1) *ML-Enabled Object Recognition*: Object recognition in AASs is enhanced by ML, especially DL models that address challenges such as occlusions, varying lighting, and movement, enabling real-time feature extraction and accurate tracking in dynamic environments. The authors in [68] introduce a method for UAVs that employs the LSTM and single-shot detectors for accurate multi-object tracking and dynamic object recognition, leveraging densely connected convolutional network (DenseNet) and YOLOv4eff networks for robust feature extraction. UGVs face the challenge of recognizing tiny objects in low-resolution scenarios, which is addressed by the generative adversarial network (GAN) proposed in [69], generating high-quality image representations that surpass traditional methods reliant on super-resolution or multi-scale features. Furthermore, USVs improve perception through the enhanced ShuffleNet and context attention feature pyramid network presented in [70], offering lightweight and efficient solutions that adapt rapidly to new categories. In [71], the authors use a variational

TABLE IV
BASIC ML'S APPLICATIONS IN AASS

Work	Category	Object	ML Method
[68]	Object Recognition	UAV	LSTM, DenseNet, YOLOv4eff
[69]	Object Recognition	UGV	GAN
[70]	Object Recognition	USV	ShuffleNet
[71]	Object Recognition	UUV	VAE
[72]	Dynamic Control	UAV	DRL
[73]	Dynamic Control	UUV	RBF
[74]	Dynamic Control	UGV	GRU
[75]	Dynamic Control	USV	DRL
[76]	Trajectory Planning	UAV	SNN, DRL
[77]	Trajectory Planning	UGVs	SVM
[78]	Trajectory Planning	USV	DDQN
[79]	Trajectory Planning	UUV	HDQN
[80]	Task Allocation	USV	SOM
[81]	Task Allocation	USV	K-means, SOM
[82]	Task Allocation	UAV/UGV	DRL
[83]	Task Allocation	UUV	BINN, SOM
[84]	Localization and Navigation	UAV	DDPG
[85]	Localization and Navigation	UGV	CNN, GRU
[86]	Localization and Navigation	USV	DDQN
[87]	Localization and Navigation	UUV	RNN
[88]	Information Fusion	UGV	CNN
[89]	Information Fusion	USV	ResNet
[90]	Information Fusion	UUV	Encoder-Decoder, attention
[91]	Information Fusion	USV	YOLOX-s, TL
[92]	Communication and Networking	UAV	GCN, meta-learning
[93]	Communication and Networking	UAV	Q-learning
[94]	Communication and Networking	UGV	LSTM
[64]	Communication and Networking	UUV	A3C
[95]	Formation Control	AAS	RL
[96]	Formation Control	AAS	DRL, IL, LSTM, TL
[97]	Formation Control	USV	DRL, RBF
[98]	Formation Control	UAV	RNN
[99]	Confrontation	UGV	DRL
[100]	Confrontation	UUV	DRL, CCN

autoencoder (VAE) network to maintain effective image representation despite changes in viewpoint and textureless images, reducing drift errors in the SLAM tasks for UUVs.

2) *ML-Enabled Dynamic Control*: ML empowers dynamic control strategies by enabling autonomous agents to autonomously adapt their decision-making and control policies in real-time, overcoming the challenge of managing environmental uncertainty and non-linearity, which traditional methods struggle to address. For instance, the authors in [72] introduce an incremental RL algorithm for UAV tracking control that transforms the task into a Markov decision process (MDP), facilitating better exploration and control policy adjustment. Moreover, for UUVs, a robust adaptive NN control scheme presented in [73] uses radial basis function (RBF) network and command-filtered backstepping to maintain tracking accuracy within preset tolerance bands despite uncertainties. UGVs benefit from the learning-and-tube-based robust model predictive control strategy investigated in [74], which utilizes gated recurrent units (GRUs) to predict state changes accurately. Similarly, USVs achieve enhance performance from the RL-based optimal tracking control detailed in [75], which addresses input nonlinearities and disturbances through NN approximators and an actor-critic framework.

3) *ML-Enabled Trajectory Planning*: ML revolutionizes trajectory planning by allowing autonomous agents to optimize paths dynamically, considering real-time variables like energy

efficiency and obstacle avoidance, and adapting to constantly changing environmental conditions with data-driven models that traditional methods cannot achieve. For instance, in [76], the authors introduce a spiking reinforcement trajectory planning approach for UAVs, which effectively minimizes energy consumption by integrating spiking neural networks (SNN) with DRL, leading to more efficient training and better energy management. Similarly, the authors in [77] propose a two-step trajectory planning method for UGVs, which combines support vector machines (SVM) for environment classification with an A* algorithm for precise path and velocity planning, offers superior adaptability to dynamic environments. For USV-enabled marine data collection, a target-oriented double deep Q-learning (DDQN) network is proposed in [78], which optimizes trajectory planning to balance energy consumption and collision avoidance, outperforming traditional methods through its ability to learn and adapt to varying conditions. Furthermore, UUVs enhance their 3D path planning capabilities with the hierarchical deep Q network (HDQN) introduced in [79], which reduces state space dimensionality, enabling more effective and efficient path planning compared to conventional approaches.

4) *ML-Enabled Task Allocation*: ML enhances task allocation by providing a flexible, adaptive framework that dynamically adjusts task distribution based on agent capabilities, environmental factors, and real-time data, addressing

critical issues such as energy efficiency, communication constraints, and scalability in complex environments. For example, the authors in [80] propose a new intelligent multi-task allocation and path planning algorithm based on the self-organizing map (SOM) neural networks and the fast marching method. This algorithm addresses two critical issues: energy consumption and communication range. It introduces a novel energy coordination scheme and a task prioritization method to efficiently allocate tasks within a USV swarm. Furthermore, the study in [81] segments the task management process into allocation and execution phases. It introduces an innovative unsupervised learning approach, utilizing an enhanced K-means algorithm for task distribution among multiple USVs, followed by the application of a SOM for task execution. Uniquely, this study also addresses the critical communication challenges inherent in constrained maritime environments by developing a competitive strategy for the K-means algorithm, considering essential factors that affect communication efficiency. As presented in [82], the authors investigate an imitation augmented DRL methods for task allocation, which facilitates complementary and cooperative coalitions in hybrid UAV and UGV swarm. Furthermore, an integrated biologically inspired SOM algorithm is described in [83] for task allocation of UUVs in complex 3D underwater environments, which combines a biologically inspired neural network (BINN) with SOM neural networks to allocate tasks efficiently.

5) *ML-Enabled Localization and Navigation*: ML transforms localization and navigation by allowing autonomous agents to learn from sensor data and environmental feedback, enabling them to maintain accurate positioning and navigation even in the face of GPS signal loss, sensor noise, and complex terrain—challenges that traditional methods fail to handle effectively. For UAVs, the state-decomposition deep deterministic policy gradient (DDPG) algorithm described in [84] separates perception-related and target-related states, thus enhancing actor network performance. Similarly, UGVs facing GPS outages benefit from the method introduced in [85], which combines convolutional neural networks (CNN) and GRUs to extract spatial and temporal features from IMU signals, thereby ensuring reliable navigation. Additionally, a robust localization algorithm using DDQN with action memory for USVs is described in [86], which integrates inertial navigation system (INS) and GPS data for precise localization. Furthermore, as presented in [87], the authors treat the UUV's navigation issue as a deep sequential learning problem and processing raw sensor data with recurrent neural networks (RNN) and attention mechanisms to avoid model building and state estimation errors.

6) *ML-Enabled Information Fusion*: ML-based information fusion enables autonomous agents to integrate diverse and imperfect sensor data, effectively enhancing environmental awareness, reducing data uncertainty, and improving decision-making, which is crucial for handling complex, unstructured environments with incomplete or noisy data. For UGVs, the end-to-end learning framework presented in [88] fuses raw camera pixels and LiDAR depth measurements, using CNNs to ensure reliable steering

command prediction even in the presence of sensor failures. The authors in [89] propose a DNN-based 3D object detection for USVs, integrating a modified proposal generation network and ResNet-based deep fusion detection network to improve feature extraction and fusion performance. Additionally, for UUVs, the hierarchical attention-based multi-source data fusion method described in [90] employs an encoder-decoder network and a fusion network with attention mechanisms to enhance fault recognition and feature orthogonalization. As presented in [91], the authors propose to employ lightweight YOLOX-s networks with TL to improve vessel detection and data augmentation for USVs.

7) *ML-Enabled Communication and Networking*: ML enhances communication and networking in AASs by enabling adaptive routing and resource allocation, optimizing network topology, and improving resilience to dynamic disruptions, thereby overcoming the limitations of static, predefined communication protocols and boosting system efficiency in real-time. To address the self-healing of UAV swarm networks post-unpredictable external destructions (UEDs), the authors in [92] develop a dual approach to rapidly restore communication connectivity using graph convolutional neural networks (GCNs). For isolated UED incidents, a GCN models recovery topologies in real-time, while for recurring UEDs, a GCN-driven trajectory planning algorithm guides UAVs to reestablish connectivity during self-healing. In [93], the authors introduce a Q-learning-based topology-aware routing protocol that enhances route reliability by incorporating two-hop neighbor information, thereby broadening the network's local view and adaptively modifying routes based on real-time conditions. Additionally, a meta-learning framework enhances GCN performance and expedites real-time operations. For secure UGV communications, the key generation scheme based on received signal strength described in [94] utilizes LSTM networks and Kalman filtering, ensuring secure communication without relying on a trusted third party. Additionally, a UUV swarm based multi-tier underwater computing framework is proposed in [64], which supports the diverse needs of the Internet of underwater things (IoUT) via asynchronous advantage actor-critic (A3C) algorithms, optimizing computing, communication, and storage resources across various platforms.

8) *ML-Enabled Formation Control*: ML elevates formation control by allowing multi-agent systems to learn optimal coordination strategies in dynamic and uncertain environments, enabling precise collision avoidance and formation adjustment without relying on predefined models, which is a critical limitation of traditional control methods. For instance, an RL-based robust control strategy for heterogeneous AASs is presented in [95], achieving optimal and collision-free formations through distributed adaptive observers and off-policy RL algorithms. Furthermore, the authors in [96] tackle the formation control with collision avoidance issue in AAS with leader-follower architecture through an DRL approach. They devise a dual-phase training strategy, initially utilizing IL with model-guided techniques for rapid proficiency gains, followed by DRL to refine the responses under varying dynamics. Their methodology integrates a formation-specific

network architecture, enhanced by LSTM for dynamic obstacle awareness and a TL scheme to boost adaptability across different environments. Moreover, for underactuated USVs, the distributed DRL approach described in [97] utilizes RBF networks to approximate hydrodynamics and unknown external disturbances, ensuring effective path following and formation control. Additionally, the study in [98] introduces an intelligent leader-follower formation control strategy based on RNNs. The dynamic model of each autonomous agent is characterized by a set of multivariate fourth-order state equations. The system is modeled as a leader-follower network with a virtual leader through directed graph theory. An adaptive formation control approach is proposed, which employs RNNs to continuously learn the system's uncertainties. By integrating consensus tracking and Lyapunov stability theories, the approach ensures coordinated flight among all followers in the formation.

9) *ML-Enabled Confrontation*: ML enhances confrontation strategies by enabling autonomous agents to rapidly adapt to adversarial situations, learn from real-time interactions, and make data-driven tactical decisions, thus improving operational effectiveness and collaboration in complex, unpredictable conflict scenarios. In UGV confrontations, the progressive DRL algorithm is introduced in [99], which optimizes autonomous maneuvering decisions using a curriculum-based training approach and self-play for iterative policy optimization. For multi-UUV attack-defense missions, the authors in [100] introduce the coding-convolutional network (CCN) to DRL to enhance cooperative decision-making with a centralized architecture and asynchronous training.

B. Requirements for Deploying DML in AAS

To realize precise decision-making and environmental adaptation, each autonomous agent needs to utilize the vast amount of data and historical information collected from various sources to enhance its ML model's performance, or collaborate with multiple agents to achieve more powerful decision-making and analytical capabilities. However, due to the resource constraints and dispersed data characteristics of AAS, centralized learning is impractical. Specifically, the reasons can be summarized as follows:

- **Data Dispersion Characteristics.** In an AAS, data are independently collected by autonomous agents, and the wireless communication links between them suffer from limited spectrum and interference. Centralizing data would lead to communication overhead, latency, bottlenecks, and reduced robustness and security.
- **Inter-Agent Correlation.** In an AAS, autonomous agents collaborate within the same environment, and each autonomous agent's position, state, cognition, and decisions affect others. Consequently, simply aggregating data to train a homogeneous model may be ineffective or even counterproductive.
- **Timely Evolution Requirements.** Each autonomous agent requires continuous evolution to adapt to dynamically changing environments, tasks, and internal states. These changes might not be captured in previously

trained data, necessitating constant model adaptation based on new data. This requirement is challenging to meet with centralized ML approaches.

Therefore, efficient and practical DML is particularly crucial for AAS. Although DML is naturally suitable for the AAS, there are still some constraints on its practical deployment. Before we introduce the use of FL and MARL on AAS, the requirements for DML's deployment in AAS are summarized as follows:

- **Model Size.** The increasing complexity of ML algorithms has led to larger models with millions of parameters. In AAS, the limited memory and computational capacity of each agent necessitate the careful management of model size to ensure that agents can efficiently store and process the required data while maintaining high performance.
- **Communication Overhead.** DML involves frequent communication between autonomous agents, requiring efficient management of both the quantity and frequency of data exchanges. Minimizing communication overhead is crucial to optimize the use of limited communication resources within the swarm. The key factors influencing this overhead are the model size and the number of communication iterations.
- **Time Cost.** The rapid execution of DML training processes is often critical, especially in time-sensitive applications. The time cost is determined by the number of iterations and the computational complexity of each iteration, which in turn influences the speed of model convergence and task completion.
- **Energy Cost.** Energy consumption is a vital concern for autonomous agents with limited power resources, especially when compared to cloud-based systems with continuous power supplies. The energy cost is influenced by factors such as model size, available computational resources, and transmission power, making energy-efficient strategies essential in AAS.

Lessons Learned: This section underscores the growing value of ML in enhancing perception, control, planning, and collaboration capabilities of AASs. However, the deployment of ML models in real-world AASs presents fundamental challenges that extend beyond algorithm design, arising primarily from the mismatch between standard ML assumptions and operational realities. Most ML methods are developed under assumptions of centralized data, stable communication, and ample computational resources—conditions that are rarely met in AAS environments. Homogeneous swarms (e.g., UAV formations) primarily encounter challenges related to scalability, synchronization, and modeling of collective behaviors. When the swarm becomes heterogeneous, these issues are further compounded by additional complexities, including role asymmetry, cross-platform data inconsistency, and incompatible control mechanisms. Directly transferring centralized ML solutions into such systems often leads to degraded performance or infeasible deployment. Although distributed learning paradigms such as FL and MARL are structurally well-aligned with AAS architectures, their practical adoption still requires careful adaptation to platform-specific constraints, including communication bandwidth,

processing power, and energy consumption. Therefore, future DML designs must embed system-awareness and resource adaptivity from the outset to ensure robust, convergent, and deployable learning under decentralized, dynamic, and resource-constrained conditions.

V. FEDERATED LEARNING FOR AUTONOMOUS AGENT SWARM

A. Overview

FL is a novel DML framework that allows multi-party collaborative ML training without sharing raw data. In basic FL, as shown in Fig. 6, a central autonomous agent coordinates a set of U autonomous agents. Each autonomous agent u possesses N_u collected data samples, denoted by $\mathcal{D}_u$. These autonomous agents independently train their models (represented by w_u) in parallel and compute their local parameters (e.g., gradients, model parameters, or variations thereof) by minimizing their loss function

$$F_u(w) = \frac{1}{N_u} \sum_{v=1}^{N_u} f(w; x_{u,v}, y_{u,v}), \quad \forall u \in \mathcal{U}, \quad (24)$$

where $f(w; x_{u,v}, y_{u,v})$ is the loss function depending on the specific ML task. These local parameters are then transmitted to the central agent. Upon receiving these local parameters, the central autonomous agent aggregates them to update the global model, denoted by w . Taking the classical FedAvg [101] as an example, the update process is given by

$$w = \frac{1}{\sum_{u \in \mathcal{U}} N_u} \sum_{u=1}^U N_u w_u. \quad (25)$$

Finally, the central autonomous agent broadcasts the updated global model back to the autonomous agents for the next iteration. Actually, the goal of FL is to find an optimal model that minimizes the global loss function as

$$w^* = \arg \min F(w), \quad (26)$$

where $w^* = w_1 = \dots = w_U$.

Compared with traditional DML frameworks, FL can support the majority of ML algorithms with satisfactory convergence speeds. Moreover, it has excellent security properties since the learning process only shares parameter information (e.g., gradients) without sharing the raw data. FL is particularly well-suited for AASs due to its low communication overhead. In general, existing FL frameworks can be classified into:

- **Horizontal FL (HFL)**, also known as sample-based FL, involves all agents collaboratively training a global ML model on their datasets. Specifically, these data samples have the same feature space but different sample spaces. In AAS applications, an example is vision-based object recognition. In this scenario, different agents observe target image data from various angles and positions and use HFL for collaborative model training or updating, thereby enhancing the accuracy of target recognition.
- **Vertical FL (VFL)**, also known as feature-based FL, differs from HFL in that the data samples of different agents share the same sample space but have different feature

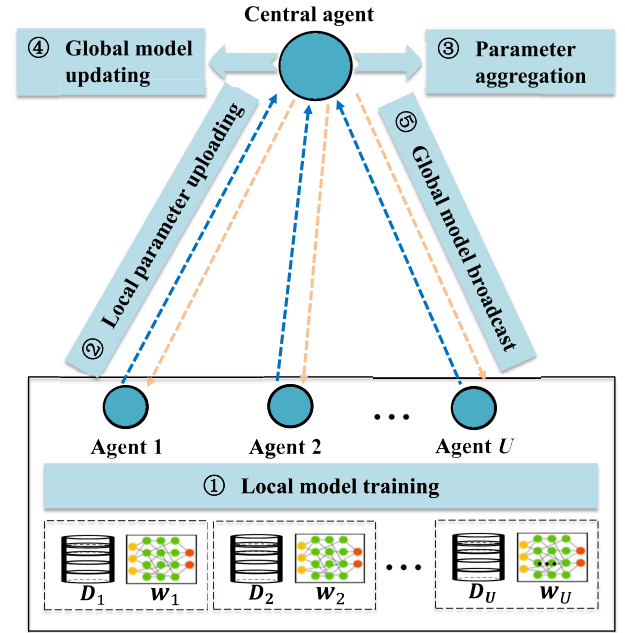


Fig. 6. The architecture of FL in AAS.

spaces. An example in AAS applications is collaborative environmental awareness, especially for heterogeneous AASs. In this scenario, each agent is equipped with different types of sensors to collect various environmental data from the same area. VFL can integrate these data to form a comprehensive environmental monitoring model, enhancing the precision and predictive capability of environmental awareness in AASs.

- **Federated Transfer Learning (FTL)** can be seen as the integration of both HFL and VFL. FTL extends the capabilities of VFL by including additional agents with datasets that possess different sample spaces and feature spaces. FTL transfers features from these diverse feature spaces into a unified representation. This unified representation is then utilized to train ML models on the aggregated data from multiple agents, enhancing the model's capability by leveraging the diversity of the datasets. By inheriting the beneficial characteristics of TL, participants in FTL can maintain their own feature spaces without needing all participants to have or use the same feature data. This makes FTL suitable for a wider range of application scenarios, especially for the task execution of heterogeneous AASs.

B. Use Cases in AAS

This section presents several classical use cases of FL in AASs, which are summarized in Table V.

1) *FL-Enabled Cooperative Perception*: FL-based collaborative perception has proven to be highly valuable in AASs and has been widely studied. For instance, the authors in [102] propose an FL-based aerial-ground sensing framework using a UAV swarm for air quality index prediction, where a dense-mobileNet model processes UAV-captured images, and

TABLE V
CLASSICAL USE CASES OF FL IN AASS

Category	Work	Object	Research Content
FL-Enabled Cooperative Perception	[102]	UAV	Aerial-ground sensing framework for air quality prediction using FL.
	[103]	UAV	FL and IL integration for UAV swarm synchronization in traffic monitoring.
	[104]	UGV	Aggregation of local features for object detection in LiDAR point clouds using FL.
	[105]	UAV	FL-based CNN and GAN approach for land cover classification.
	[106]	UAV	Semi-supervised FL framework for UAV image recognition.
FL-Enabled Cooperative Localization and Navigation	[107]	UAV	Federated RL algorithm for heterogeneous UAV swarm navigation.
	[108]	AAS	FL and DQN integration for security-preserving navigation.
	[109]	UAV	FL framework for localization with improved accuracy and robustness.
	[110]	UAV	Fault-tolerant localization solution combining model-based and data-driven techniques.
FL-Enabled Cooperative Communication	[111]	UGV	FL for accelerating sector selection in mmWave bands.
	[112]	UGV	FL-aided joint power control and power allocation strategy in UGV networks.
	[113]	UGV	Federated multi-agent DRL approach for power control and channel selection.
	[114]	USV	FL-based secure communication scheme for USVs in maritime IoT networks.
FL-Enabled Cooperative SLAM	[115]	AAS	FL for real-time feature extraction and dynamic vocabulary creation.
	[116]	UGV	FL-based dynamic map fusion framework for accurate object prediction.
	[117]	UGV	FL and RHS-based SLAM system.
	[118]	UGV	FL-based deep unrolling method for LiDAR super-resolution.
FL-Enabled Cooperative Control	[119]	UAV	FL-based MFG approach for controlling large UAV populations.
	[120]	UGV	FL and RL integration for UGV trajectory control, improving trajectory tracking.
	[121]	UGV	Dynamic FL proximal algorithm for UGV control considering wireless characteristics.
FL-Enabled Security	[122]	UAV	FL-based security architecture for jamming attack detection in UAV swarms.
	[123]	UAV	Intrusion detection system using FL and CGAN, enhanced with LSTM networks.
	[124]	UAV	FL-based UAV authentication method using radio frequency features.
	[125]	UGV	Trust management method combining FL and blockchain for secure model storage.

a graph convolutional neural network-based LSTM (GC-LSTM) model analyzes spatiotemporal data on the ground. Furthermore, [103] explores the combination of FL and IL to synchronize UAV swarm actions in cooperative traffic monitoring. This approach leverages generative adversarial IL and self-IL to enhance performance despite isolated environments. The study in [104] addresses the challenge of object detection in heterogeneous LiDAR point clouds from UGVs by aggregating local features across multiple UGVs, thereby ensuring adaptability across different scenarios. Similarly, the authors in [105] extend the use of FL to land cover and land use classification, employing CNNs on high-resolution UAV imagery and integrating GANs to counter data scarcity and prevent overfitting, as show in Fig. 7. Considering the limitations of supervised learning, the work in [106] introduces a semi-supervised FL framework for UAV image recognition, with the flexible aggregation rule specifically designed to manage statistical heterogeneity.

2) *FL-Enabled Cooperative Localization and Navigation*: Ensuring accurate and safe localization and navigation of AASs in unknown environments is critical. FL leverages dispersed data to enhance positioning accuracy and navigation for autonomous agents while providing robust security guarantees, leading to extensive research on FL-enabled localization and navigation. The authors in [107] introduce a decentralized framework for heterogeneous UAV swarm navigation. This framework features a novel soft hierarchical DRL network that learns a generalized upper-level navigation policy, aimed at minimizing discrepancies among the policy networks of different UAV types. Moreover, a distinct federated RL algorithm is proposed, enhancing knowledge sharing among

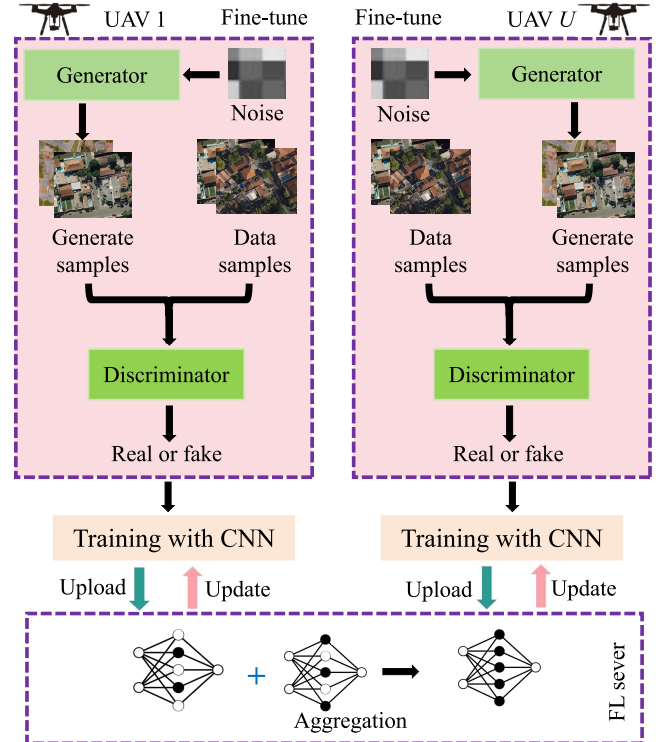


Fig. 7. A scenario of FL-enabled cooperative perception.

heterogeneous UAVs by aggregating similar policy models on the server side and executing adaptive gradient aggregation locally on UAVs, thus facilitating the development of a more universal navigation policy. Regarding security, the

work in [108] combines deep Q-network (DQN) with FL to ensure security-preserving navigation of mobile robots, ensuring privacy without sacrificing accuracy. Similarly, the study in [109] propose an FL framework that employs a state-space model for single-agent localization and the alternating direction multiplier method for multi-agent scenarios, which shows excellent transfer performance in terms of accuracy and robustness. Furthermore, a fault-tolerant localization solution is introduced in [110] that combines model-based and data-driven techniques within an FL framework, ensuring accuracy even in the presence of sensor faults.

3) *FL-Enabled Cooperative Communication*: Due to its ability to efficiently leverage decentralized data for ML model generation and its security advantages, FL has become a popular research focus in cooperative communication. Specifically, fast sector-steering in mmWave bands for UGV mobility is challenging due to the lengthy exhaustive search over predefined antenna sectors, which is impractical for short contact times. Hence the authors in [111] propose using FL to accelerate sector selection by integrating data from non-RF sensors like LiDAR, GPS, and cameras. They introduce a multi-modal FL architecture that fuses inputs from these sensors to predict optimal antenna sectors for vehicle alignment. To tackle the joint power and resource allocation challenge for ultra-reliable low-latency communication in UGV networks, the authors in [112] introduce a novel reliability measure based on extreme value theory to handle extreme queue length events, and consequently to minimize power consumption while ensuring high reliability. Similar to [111], an FL-based prediction approach is proposed to estimate the tail distribution of queue lengths, assuming they are independently and identically distributed among vehicular users. Furthermore, to cope with the issues of dynamic topology, fast-changing channels, and time sensitivity in cellular-underlying UGV communications, a federated multi-agent DRL approach fully integrating the advantages of both RL and FL is proposed in [113] to optimize channel selection and power control in a decentralized manner for satisfying reliability and delay requirements while maximizing cellular link transmit rates. For security considerations in USV swarm-assisted maritime IoT networks, the authors in [114] propose an FL-based communication and computation scheme, where USVs transmit trained models to a high-altitude platform (HAP) using nonorthogonal multiple access (NOMA) for global aggregation. To prevent eavesdropping, chaotic sequences are used to spread model information before broadcasting. Their approach aims to minimize energy consumption by optimizing global and local accuracy, HAP transmission power, and NOMA transmission duration while ensuring secrecy and meeting latency constraints.

4) *FL-Enabled Cooperative SLAM*: SLAM among agents not only extends sensing ranges but also improves accuracy, making it a critical component in AASs. FL can significantly enhance the performance of SLAM by utilizing data collected from various agents. For instance, the authors in [115] propose a cloud robotic SLAM architecture utilizing FL, which enhances performance through an FL algorithm designed for real-time feature extraction and dynamic vocabulary creation.

This approach ensures that image data remains localized to agents, thus preserving privacy and reducing data transmission overhead. Building on this concept, the authors in [116] introduce an FL-based dynamic map fusion framework that maintains high map quality, even in the face of unknown object counts, sensing uncertainties, and missing data labels. Their framework includes a three-stage fusion scheme for accurate object prediction, local map fusion using fidelity scores, and a knowledge distillation (KD) method to generate training labels when data labels are unavailable. In a related advancement, the work presented in [117] presents an innovative SLAM system that leverages reconfigurable holographic surfaces (RHS) and FL, significantly reducing hardware costs by replacing expensive phased array antennas with affordable RHS metasurfaces. This FL framework facilitates collaboration among multiple UGVs, improving SLAM performance by integrating additional sensed data. To manage the complexities introduced by RHS's unique radiation structure and federated information exchange, a cooperative SLAM protocol is proposed, governing RHS-based radar sensing and data processing across vehicles. Further enhancing SLAM accuracy, the authors in [118] propose a federated deep unrolling method to achieve LiDAR super-resolution while minimizing the costs associated with high-resolution sensors. This approach involves multiple UGVs collaborating with a server to learn a neural network regularizer that captures LiDAR data dependencies, transforming the solution into a DL architecture with explainable parameters. The method incorporates FL with an adapt-then-combine strategy, enabling each agent to optimize its model and merge regularizers into a robust global version, thereby improving the overall SLAM performance.

5) *FL-Enabled Cooperative Control*: Real-time control of numerous autonomous agents is critical for mission-critical tasks. To tackle the challenge of controlling large UAV populations, the authors in [119] propose an FL-based mean field game (MFG) approach. Traditional methods relying on extensive inter-UAV communication are impractical for real-time applications. The MFG framework reduces communication needs but requires powerful processors. To address this, neural networks approximate solutions to the Hamilton-Jacobi-Bellman and Fokker-Planck-Kolmogorov equations. FL then shares these neural network model parameters among drones, ensuring the convergence conditions of the MFG are met. The study in [120] combines FL and RL to enhance UGV trajectory control. This strategy leverages the parallel learning capabilities of UGVs, which operate as intelligent agents and communicate their findings to a central system running at the edge. Each UGV includes modules for state estimation, rewards calculation, action selection, and policy update. The federated server aggregates individual UGV policies into a group policy using a learning aggregation function. The approach demonstrates significant improvements in trajectory tracking over traditional controllers, with notable reductions in mean absolute error, root mean square error, and standard deviation. Additionally, the proposed method accelerated learning and outperformed optimized fuzzy logic controllers across various trajectories, such as ellipse and octagon. Building

on this, considering the impact of wireless connectivity on accurate control decisions in dynamic conditions, the authors in [121] introduce a dynamic federated proximal algorithm that considers UGV mobility, wireless channel dynamics, and non-IID data. The algorithm's convergence is rigorously analyzed under realistic conditions, highlighting the impact of variable UGV participation and data quality.

6) *FL-Enabled Security*: AASs are susceptible to malicious attacks like jamming, which disrupt communications by interfering with receiver signals. Detecting jamming attacks in AASs is particularly challenging due to the decentralized nature of the network, the agent's mobility, and varying communication ranges. Centralized detection is impractical, and existing decentralized solutions do not account for unbalanced sensory data in dynamic environments. The authors in [122] propose an FL-based security architecture for jamming attack detection in UAV swarms. This architecture enhances detection by incorporating a agent group prioritization technique using the Dempster-Shafer theory, addressing unbalanced data properties and communication efficiency. Similarly, the authors in [123] develop an intrusion detection system using a conditional GAN (CGAN) combined with blockchain-supported FL. Integrating LSTM networks enhances CGAN's data generation, improving intrusion detection accuracy. Differential privacy (DP) and blockchain technology ensure secure and private collaborative model training and updating, significantly enhancing detection capabilities. Additionally, UAV authentication is an important issue that needs to be addressed. The study in [124] introduces an FL-based UAV authentication approach leveraging UAVs' radio frequency features. This approach employs a locally optimized DNN architecture on UAVs using stochastic gradient descent. Additionally, homomorphic encryption (HE) and secure aggregation are used to protect model parameters. Similarly, in UGV swarm, the authors in [125] propose an FL and blockchain-based method for trust management, which involves a trust evaluation to refine the data used in the FL process and a blockchain-based reputation system to secure the storage and sharing of global ML models.

C. Key Techniques

The key techniques of FL applied to AASs mainly focus on aggregation mechanisms, communication compression, resource management, and data improvement. Table VI summarizes the key techniques in FL applied to AASs.

1) *Aggregation Mechanisms*: In FL, there is a need to continuously aggregate the local updates to update the global model. The existing aggregation mechanisms can be classified into three types: synchronous aggregation [126], [127], [128], [129], [130], [131], asynchronous aggregation [132], [133], [134], [135], and hierarchical aggregation [136], [137], [138], [139], [140], [141].

- **Synchronous Aggregation**. Synchronous aggregation is widely adopted in FL frameworks such as FedAvg and FedSGD [101], where the central node waits for all participating agents to upload their local updates before performing global aggregation. While this design ensures

deterministic synchronization, it is also sensitive to straggler agents, especially in dynamic or resource-constrained environments. To improve the generalization performance under synchronous settings, [126] explores the effect of shrinking the sum of aggregation weights (i.e., making it less than 1), which acts similarly to weight decay. This regularization effect is shown to depend on both data heterogeneity and the number of local training epochs. In addition, several works focus on heterogeneity-aware aggregation. For example, [127] proposes a parameter-wise elastic weighted averaging strategy, where each agent quantifies the importance of its internal parameters and contributes them accordingly during aggregation. Similarly, [128] incorporates both local dataset sizes and category distribution divergence into the weight computation, achieving tighter convergence bounds and improving model utility. Personalized and adaptive aggregation has also gained traction. The "learning to collaborate" framework in [129] optimizes agent-specific mixing weights and employs attention-based mechanisms to handle new or unseen agents. Reference [130] introduces dynamic weight adjustment to personalize the aggregation process by estimating inter-agent similarities based on submitted models. Further, [131] proposes cross-round aggregation, which selectively integrates historical local models that are most consistent with others to reduce global-local divergence. Synchronous aggregation is particularly well-suited for small- to medium-scale AAS swarms with well-aligned task cycles, such as UAV formations engaged in synchronized search or periodic patrols. In such scenarios, the central controller can define aggregation windows that accommodate most agents' reporting latencies. To mitigate the impact of occasional stragglers without compromising synchronization, delay-tolerant techniques—such as reusing previous-round updates or using predictive interpolation—can be applied. These adaptations help maintain model convergence and continuity while preserving the advantages of synchronous control, which aligns well with the structured mission execution typically observed in AAS deployments.

- **Asynchronous Aggregation**. Asynchronous aggregation has attracted growing attention in FL, especially in the context of AASs, where unreliable communication, dynamic system states, and resource heterogeneity can lead to frequent delays and poor scalability. Unlike synchronous approaches, asynchronous aggregation allows the central node to update the global model immediately upon receiving local updates from any participating agent, without waiting for others. This decouples agent operations and reduces idle time, making it suitable for non-uniform participation scenarios. To address low-efficiency and unreliable nodes, [132] proposes a lightweight asynchronous scheme that selectively controls both update frequency and parameter participation. Additionally, they incorporate a mutual reliability evaluation mechanism, enabling the detection and mitigation of model poisoning by adjusting aggregation weights

TABLE VI
SUMMARY OF KEY TECHNIQUES IN FL APPLIED TO AASS

Category	Subcategory	Key methods
Aggregation mechanisms	Synchronous aggregation	Aggregation weight shrinking [126]; Parameter-wise elastic averaging [127]; Dataset size and category distribution-based aggregation [128]; Attention-based weighted aggregation [129]; Dynamic weight adjustment [130]; Selective cross-round aggregation [131]
	Asynchronous aggregation	Lightweight asynchronous aggregation [132]; Scheduling policy with channel and data representation [133]; Adaptive asynchronous aggregation [134]; Dynamic asynchronous aggregation [135]
	Hierarchical aggregation	DRL-based staleness control [136]; Reputation-based aggregation [137]; Multi-layer blockchain-enabled aggregation [138]; Decentralized aggregation with gossip steps [139]; Adaptive layer-wise decentralized aggregation [140]; DAG-enabled decentralized aggregation [141]
Communication compression	Quantization	Quantized stochastic gradient descent [142]; Universal vector quantization [143]; Dynamic adaptive quantization [144]; Layer-based quantization with error feedback [145]; Channel quality-aware quantization [146]
	Sparsification	Aggressive gradient sparsification [147]; Adaptive gradient sparsification [148]; Time-correlated sparsification [149]; Lossless gradient sparsification [150]; Joint sparsification-quantization [151]
	KD	Group knowledge transfer training [152]; Adaptive mutual KD [153]; Distributed GAN-based KD [154]
Resource management	Resource scheduling	Convergence gap-minimization via agent selection, transmission power control and bandwidth allocation [155]; Time-minimization via agent selection and bandwidth allocation [156]; Lyapunov-based online agent selection and bandwidth allocation [157]; Agent selection and beamforming design [158]; Joint communication and computation scheduling [159]; Joint caching decision, agent selection, and resource allocation [160]
	Joint learning and resource co-design	Joint batch size selection and resource allocation [161]; Joint local iteration accuracy configuration and resource allocation [162]; Joint agent selection and sparsification [163]; Dynamic agent selection, quantization and resource allocation [164]; Over-the-air computation-based FL [166]; Over-the-air computation and 1-bit compression sensing based FL [165]
	Joint movement, learning, and resource co-design	Joint UAV deployment and resource allocation [167]; Joint UAV deployment, sensing, computation, and communication resource allocation [168]; Joint learning, UAV's trajectory design and resource allocation [169]; Joint learning, UAV's trajectory design and resource allocation in mmWave MIMO UAV networks [170]
Data improvement	Data filtering	Systematic data sample selection [171]; Data sample selection with adaptive deadline control [172]; Joint agent and sample selection [173]; ADMM and K-means based data filtering [174]
	Data augmentation	CVAE-based data augmentation [175]; Selective GAN-based data augmentation [176]; Distributed GAN-based data augmentation [177]; Conditional denoising diffusion-based data augmentation [178]

based on peer assessments. Tackling the issue of aggregation bias and variance, [133] explores a scheduling policy that accounts for both communication channel quality and the representativeness of each agent's training data, ensuring more balanced update integration. Adapting to dynamic AAS environments, [134] proposes a resource-aware aggregation mechanism that adjusts local training workloads according to real-time resource availability at each agent. This increases participation rates and reduces update staleness, ultimately improving convergence speed and accuracy. Similarly, [135] designs a dynamic weighting strategy that considers both the staleness of updates and local loss impact, enabling the global model to more effectively incorporate asynchronous contributions. Asynchronous aggregation aligns naturally with the operational characteristics of AASs deployed in challenging environments—such as long-duration underwater monitoring by UUVs, wide-area terrain mapping by UAVs, or offshore patrols by USVs—where intermittent connectivity, mobility-induced delays, and heterogeneous sensing cycles are common. In such settings, asynchronous FL allows agents to upload their updates flexibly, based on local task progress and resource availability. To ensure training robustness, the central node can incorporate staleness-aware weighting schemes, prioritize contributions from agents covering critical or under-sampled regions, and dynamically modulate model integration strategies to balance recentness and relevance.

These adaptations allow learning to proceed smoothly despite asynchronous behavior, enabling the global model to progressively capture spatially diverse knowledge from the swarm.

- Hierarchical Aggregation.** Both synchronous and asynchronous aggregation typically rely on a star topology, where a central agent connects directly to all participating agents. While simple, this topology suffers from limited scalability, poor robustness, and communication bottlenecks in resource-constrained environments. To address these limitations, hierarchical and decentralized aggregation mechanisms have been proposed to better suit large-scale and distributed FL scenarios. One representative approach is the hybrid hierarchical FL framework in [136], which combines synchronous agent-edge and asynchronous edge-cloud aggregation. It incorporates DRL to adaptively manage staleness and uses a heterogeneity-aware agent-to-edge association policy. Building on this, [137] introduces a secure hierarchical FL scheme that defends against poisoning attacks via a logarithmic normalization method and a reputation-based edge-level aggregation process. To enhance both latency and security, [138] integrates blockchain into a multi-layer hierarchical FL architecture. It formally analyzes how imbalanced local data distributions affect model accuracy and provides a mathematical link between agent associations and model error. Meanwhile, [139] proposes two decentralized FL

variants: one leverages sharpness-aware minimization to yield flatter and more robust local optima, while the other uses multi-round gossip communication to improve consistency and convergence without relying on a central node. Going further, [140] introduces an adaptive, layer-wise decentralized training scheme. By ranking model layers based on learning dynamics and using a max-match algorithm, it determines optimal aggregation strategies that consider both data heterogeneity and communication costs. Finally, [141] proposes IronForge, a fully decentralized FL framework that addresses central coordinator dependency and fairness via a directed acyclic graph (DAG)-based model tracking system, where model updates are treated as graph nodes and selectively referenced during aggregation. Hierarchical aggregation mechanisms are particularly well-suited for large-scale AAS deployments that are geographically distributed, such as environmental monitoring, agricultural surveillance, or ocean observation tasks. In such scenarios, a multi-tier aggregation architecture can be constructed, comprising intra-cluster edge aggregators, inter-cluster fusion layers, and a centralized cloud server. Agents within a region can first perform local aggregation—either synchronously or asynchronously—at the cluster level. These intermediate models are then integrated across clusters at edge nodes and periodically uploaded to the cloud for global model optimization. This hierarchical structure reduces the burden on long-range communication links, accommodates heterogeneous connectivity conditions, and enhances both scalability and training efficiency across multiple AASs, all while maintaining convergence guarantees.

2) *Communication Compression*: Although FL can enhance communication efficiency by avoiding the transmission of raw data, the number of parameters that need to be exchanged during the iteration process is still quite substantial as the ML models involved become increasingly complex. To further reduce the communication overhead of FL, three kinds of communication compression techniques are widely adopted, i.e., quantization [142], [143], [144], [145], [146], sparsification [147], [148], [149], [150], [151], KD [152], [153], [154].

- **Quantization.** Quantization techniques aim to reduce the communication overhead in FL by encoding local updates with fewer bits before transmission to the central node. This is particularly critical for resource-constrained agents that operate under tight bandwidth or energy budgets, such as in many AAS applications. Early work such as [142] introduces a quantized stochastic gradient descent algorithm with theoretical convergence guarantees. This method enables bit-rate tuning per iteration, allowing agents to balance communication cost and convergence rate, albeit with increased gradient variance. To further reduce distortion during model transmission, [143] adopts universal vector quantization in FL and demonstrates through theoretical analysis that quantization error diminishes as the number of agents grows. To enhance adaptivity, [144]

proposes a dynamic quantization algorithm that adjusts quantization levels both temporally and across different agents, improving compression while maintaining model performance. Similarly, [145] introduces a layer-wise quantization scheme with error feedback, where bit allocation and quantization functions are selected based on delta parameter distribution and entropy characteristics. From a theoretical standpoint, [146] analyzes the impact of quantization error, channel diversity, and aggregation strategies on FL convergence, and further formulates an optimization problem to allocate quantization bits adaptively based on channel quality to minimize convergence bounds. In AASs, quantization is especially advantageous due to agents' limited communication and power budgets. However, effective deployment requires task- and agent-aware customization. For example, UUVs relying on narrowband acoustic links may benefit from aggressive low-bit quantization schemes combined with error-feedback mechanisms to preserve learning performance under tight bandwidth. In contrast, UAVs equipped with intermittent but higher-bandwidth radio communication can adopt adaptive bit-width quantization that dynamically adjusts based on link quality, mobility, or task priority. To maintain model accuracy under these constraints, quantization-aware training and regularization techniques can be integrated to compensate for reduced precision. Overall, tailoring quantization strategies to the communication modality and operational context of each agent is critical to achieving efficient, scalable FL in real-world AAS deployments.

- **Sparsification.** Sparsification refers to selecting a subset of local update elements for transmission to the central nodes, thereby reducing communication costs in FL. In [147], the authors propose a rigorous algorithmic framework that accounts for both objective and system heterogeneity under infrequent communication, while ensuring linear convergence even under aggressive gradient sparsification. This work provides one of the first comprehensive theoretical analyses quantifying the effect of compression on convergence in FL. To further improve communication efficiency, [148] introduces an adaptive sparsification scheme that applies a two-step process: conventional sparsification followed by a refinement step to retain the most informative gradient components. In [149], a time-correlated sparsification strategy is proposed that exploits temporal correlations in sparse gradients across iterations to prioritize significant parameters. Similarly, [150] reduces communication by projecting gradients into a novel space derived from input activations and output gradients, achieving lossless sparsification with competitive or even superior performance compared to full gradient uploads. In another line of work, [151] presents a unified sparsification-quantization framework by formulating sparsity as zero-bit quantization and optimizing a mixed-precision policy to minimize compression noise. This approach achieves compression ratios up to 1000x with minimal accuracy degradation. In practical AAS scenarios, sparsification

techniques are especially beneficial for reducing communication burdens across large-scale agent swarms. For example, in UAV-based agricultural monitoring, agents can sparsely transmit updates relevant only to disease-prone or anomalous crop zones, eliminating redundant gradient transmissions. Similarly, UUV swarms conducting oceanographic surveys over low-bandwidth acoustic links can prioritize uploading updates tied to detected environmental events (e.g., temperature anomalies or biological activity), maximizing transmission utility under tight constraints.

- **KD.** KD is a model compression technique where a smaller student model learns to mimic the behavior of a larger, more accurate teacher model. In FL, KD is employed to reduce the size of transmitted models and mitigate resource constraints while preserving model quality. In [152], a group knowledge transfer framework is developed that alternates between training small local models on constrained agents and distilling their collective knowledge into a central larger model. This method supports asynchronous updates and achieves accuracy on par with FedAvg, while reducing local computation and bandwidth usage. To further enhance communication efficiency, [153] combines adaptive mutual distillation with gradient compression. Addressing privacy and input-sharing limitations, [154] proposes a generative KD framework in which a central agent hosts a generator that produces shared representations, while local agents act as distributed discriminators. This setup enables data-independent knowledge transfer and improves training convergence and scalability. KD is particularly suitable for heterogeneous AAS environments, where agents differ in computational and memory capabilities. Instead of synchronizing full models, lightweight platforms such as micro-UAVs can train compact student models and periodically distill knowledge from stronger peers or cloud-based aggregators. This enables flexible collaboration without requiring tight parameter alignment. Moreover, generative and data-free KD approaches are especially relevant in privacy-sensitive AAS tasks—such as military surveillance or ecological monitoring—where raw data sharing is restricted. Such methods facilitate asynchronous and secure model improvement across diverse agents, aligning well with the operational constraints of AASs.

3) *Resource Management:* Given the limitations and heterogeneous resources in AASs, it is essential to efficiently manage these dispersed resources to enhance FL performance while minimizing resource costs. Existing works on FL's resource management include resource scheduling [155], [156], [157], [158], [159], [160], joint learning and resource co-design [161], [162], [163], [164], [165], [166], and joint movement, learning, and resource co-design [167], [168], [169], [170].

- *Resource Scheduling.* In early stages, several works focused on optimizing the scheduling of communication and computation resources to enhance FL performance in real-world deployments. Specifically, the authors

in [155] present an early effort to demonstrate the impact of unreliable transmission on FL's convergence. This work proposes a joint optimization of participant selection, transmission power, and bandwidth allocation to eliminate the convergence gap caused by transmission errors and partial collaboration, while considering time and energy consumption constraints. Moreover, the work presented in [156] formulates a training time minimization problem by optimizing agent selection and bandwidth allocation. Furthermore, the study in [157] addresses the joint agent selection and bandwidth allocation problem from a long-term perspective, designing a Lyapunov-based online algorithm to solve the per-round optimization problem. Additionally, in [158], the authors propose a joint agent scheduling and receive beamforming design to improve FL performance in wireless multiple-input multiple-output (MIMO) networks. Beyond communication resource allocation, the work in [159] introduces computation resource optimization into the FL framework. Additionally, the work in [160] presents a knowledge caching framework in FL, where a joint optimization of caching decisions, agent selection, and resource allocation (including transmission power and bandwidth) is conceived to minimize training error bounds, subject to cache capacity, latency, and energy consumption constraints.

- *Joint Learning and Resource Co-design.* Given the limited performance improvements from merely optimizing physical resources, recent studies have shifted towards cross-layer optimization, which integrates both resource and learning mechanism optimizations. The work in [161] introduces a performance metric called learning efficiency and formulates a training optimization problem that involves joint optimization of batch size selection and communication resource allocation. Similarly, [162] proposes a framework that jointly optimizes learning and resource allocation, encompassing local iteration accuracy, time and bandwidth distribution, power control, and computation frequency adjustment. Furthermore, the studies in [163] and [164] integrate communication compression with resource configuration. Specifically, [163] examines how sparsification and agent selection impact energy consumption throughout FL's convergence process, while [164] proposes a dynamic optimization method that adjusts agent selection, gradient quantization levels, and resource allocation to minimize error bounds within given time and energy constraints over a long-term horizon. However, due to limitations of digital communication schemes, which struggle to scale with the growing number of agents—resulting in increased communication resource demands and training delays—some studies have explored over-the-air computation-enabled FL to address these issues, as seen in [165], [166]. For example, [165] presents a dynamic agent scheduling mechanism that selects suitable agents for model training with proper power control, taking into account data importance, channel conditions, and energy consumption. This approach also retains updates from unselected agents

for future use. Additionally, [166] introduces a 1-bit compressive sensing scheme for over-the-air computation in FL, and develops a joint optimization framework for agent scheduling and power scaling using the alternating direction method of multipliers to handle large-scale agent networks.

- **Joint Movement, Learning and Resource Co-design.** Considering the significant impact that the movement and spatial location of agents have on FL's training process, it becomes essential to account for these dynamic factors when optimizing system performance. In light of this, the authors in [167] address the optimization of UAV deployment and resource allocation to enhance learning performance and energy efficiency, employing successive convex approximation techniques. Additionally, the authors in [168] explore how UAV positioning influences sensing quality, proposing a probabilistic sensing model to improve data sample quality and deriving an upper bound for FL's training loss. Their study formulates a training time minimization problem by jointly optimizing UAV deployment, sensing, computation, and communication resources. However, both [167] and [168] assume static agent locations during FL's training process, which does not reflect real-world scenarios where agent positions typically change, especially during extended training periods. To address this, the authors in [169] propose a distributed proximal policy optimization-based strategy that jointly optimizes UAV's trajectory, artificial noise transmission power, CPU frequency, transmission power, bandwidth allocation, and local model accuracy to minimize FL's training time and energy consumption. Furthermore, the authors in [170] extend this optimization problem to mmWave massive MIMO UAV networks, considering factors such as UAV's trajectory, transmission power, analog beamforming, and local iteration accuracy.

4) **Data Improvement:** It is well known that the quality of collected data significantly influences the performance of ML models. In AASs, harsh environments and potential attacks often lead to suboptimal data quality. Therefore, techniques to enhance agents' data quality are crucial. Existing approaches to improve data quality in FL primarily focus on data filtering [171], [172], [173], [174] and data augmentation [175], [176], [177], [178].

- **Data Filtering.** Data quality plays a pivotal role in the effectiveness of FL, particularly in AAS environments where data is often noisy, incomplete, or even adversarially manipulated due to sensing limitations, environmental interference, and security threats. To this end, various data filtering strategies have been proposed to ensure that only high-quality and reliable data contributes to model training. One line of work focuses on identifying and excluding data samples that are erroneous, unbalanced, or lacking diversity—factors that can lead to model divergence or degraded generalization. For instance, [171] presents a comprehensive data sample selection framework that simultaneously considers label noise, feature redundancy, and distribution skewness.

Another strategy emphasizes selecting “informative” data while accounting for agent-level constraints. In [172], the authors design a utility-driven filtering method that prioritizes valuable samples, guided by agents' computational capacity and security conditions. They further incorporate an adaptive deadline prediction mechanism to optimize training efficiency under resource constraints. Going beyond static filtering, some studies explore joint agent-sample optimization to dynamically mitigate noise without discarding valuable hard examples. In [173], the proposed method integrates sample selection with agent participation, using meta-learning to identify samples that positively contribute to the model while filtering harmful ones. This approach is particularly suitable for AAS, where both agent availability and data quality can fluctuate significantly. From a security perspective, [174] addresses data poisoning risks in distributed SVMs by combining edge-cloud FL with a robust Alternating Direction Method of Multipliers (ADMM)-based filtering pipeline. This system detects compromised agents and clusters trusted samples via K-means, enhancing resilience against adversarial attacks. In AAS scenarios, traditional data filtering strategies must be significantly adapted to cope with the inherent spatial heterogeneity, sensing uncertainties, and vulnerability to attacks characteristic of such systems. For instance, in UUV swarms where acoustic sensors have high uncertainty due to reflections and interference, filtering algorithms should leverage spatial correlation by cross-validating sensor readings among neighboring agents rather than solely relying on local statistical outlier detection. Similarly, UAV swarms facing targeted data-poisoning or spoofing attacks require filtering mechanisms based on multi-agent consensus or reputation-driven adaptive sampling to dynamically exclude data from compromised sources. These context-aware filtering approaches, tailored specifically to AAS characteristics, ensure that critical but potentially subtle environment signals are preserved while effectively mitigating harmful data influences.

- **Data Augmentation.** Data augmentation is a widely adopted strategy in FL to mitigate data scarcity and heterogeneity by generating new data samples from limited existing observations. This is particularly important in AAS, where agents often operate in isolated or constrained environments, making data collection costly and sparse. To enhance model diversity and generalization, [175] proposes a CVAE-based augmentation method that applies constraints to the latent space and incorporates attention mechanisms to produce easily distinguishable cross-class samples. This method not only improves model accuracy by up to 8% but also reduces communication costs by 50%. In another approach, [176] addresses the communication burden and performance degradation caused by non-IID data through a selective GAN-based scheme. Here, representative local data is used to train GANs centrally, and the generated data is redistributed for local model calibration via a

dual-calibration strategy—achieving both efficiency and robustness. In VFL scenarios, where overlapping sample instances are limited, [177] introduces a distributed GAN with a local discriminator feedback aggregation mechanism. This approach improves each party's ability to learn local data distributions by generating plausible synthetic instances across multiple sources. Complementing these efforts, [178] develops a conditional denoising diffusion probabilistic model to generate label-aligned synthetic data at a central agent and distribute it to local agents, effectively improving personalization. However, effective data augmentation in AAS requires more than conventional image-based techniques. It must explicitly account for domain-specific constraints such as limited local observations, spatial-temporal heterogeneity, and complex environment dynamics. For instance, in UAV swarms conducting long-term environmental surveillance over heterogeneous terrains, basic transformations (e.g., rotation or cropping) may offer limited value. Instead, augmentation can integrate physics-informed models or generative diffusion frameworks to simulate rare yet mission-critical conditions—such as extreme weather or low-visibility terrain anomalies—thus enhancing model resilience. Likewise, in UUV swarms engaged in marine exploration, augmentation may involve generating synthetic acoustic signals that reflect realistic propagation effects or rare oceanographic patterns, which are otherwise difficult to capture. These task- and modality-specific augmentation strategies are essential to build robust and generalizable FL models that perform reliably under the diverse and uncertain conditions characteristic of AAS.

Lessons Learned: This section provides a task-grounded and system-aware survey of FL in AASs, highlighting not only how FL has been applied across various mission tasks—ranging from object recognition and control to navigation and collaborative confrontation—but also what system-level techniques are necessary to make these applications viable in real-world deployments. A key insight is that FL in AASs is not merely about distributing learning across agents, but rather presents a multi-faceted engineering challenge shaped by model size constraints, communication bottlenecks, computational heterogeneity, and dynamic tasking. Our review shows that relying on standard synchronous FL schemes without considering agent availability, transmission delays, or dynamic topologies often leads to suboptimal outcomes. Instead, successful FL deployment in AASs requires flexible integration of compression techniques, communication-aware scheduling, aggregation strategies tolerant to asynchrony, and architectural adaptations for partial model sharing. Notably, there is no universally optimal FL strategy; rather, the choice of mechanisms—whether aggregation style or update frequency—must be tailored to the mission type, agent roles, and operational conditions. In conclusion, enabling FL in AASs demands not just algorithmic innovation, but holistic system-level adaptation to constraints inherent to autonomous, decentralized, and mission-driven multi-agent environments.

VI. MULTI-AGENT REINFORCEMENT LEARNING FOR AUTONOMOUS AGENT SWARM

In FL, we implicitly assume that datasets across different autonomous agents remain fixed and independent. However, this assumption is not viable when agents interact within a shared environment and make decisions based on their own observations. In these scenarios, MARL offers a more nuanced representation of the interactions between agents and the environment. It is worth noting that, although previous sections have introduced some works on federated reinforcement learning, it fundamentally differs from multi-agent reinforcement learning. Federated reinforcement learning essentially distributes the training of reinforcement learning models using federated learning, with relatively weak coupling among different agents. MARL adeptly addresses the complexities of inter-agent cooperation and competition, adapting to dynamic and intricate environments, thereby providing a more robust and flexible solution [179].

A. Overview

MARL is usually modeled as Markov games (MG) (also named stochastic games or dynamic games). An MG is characterized by the tuple $(\mathcal{U}, \mathcal{S}, \{\mathcal{A}_u\}_{u \in \mathcal{U}}, \mathcal{P}, \{R_u\}_{u \in \mathcal{U}}, \gamma)$, which includes the following elements [180]:

- $\mathcal{U} = \{1, \dots, U\}$: A set of U agents, where $U > 1$.
- $\mathcal{S}$: The state space observable by all agents.
- $\mathcal{A}_u$: The action space specific to agent u .
- $\mathcal{A} = \mathcal{A}_1 \times \dots \times \mathcal{A}_U$: The joint action space formed by the actions of all agents.
- $\mathcal{P} : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$: The transition probability function, which describes the probability distribution over the next states $s' \in \mathcal{S}$ given the current state $s \in \mathcal{S}$ and joint action $a \in \mathcal{A}$.
- $\{R_u\}_{u \in \mathcal{U}}$: The set of reward functions $R_u : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$, where R_u specifies the immediate reward received by agent u when transitioning from state s to state s' due to the joint action a .
- $\gamma \in [0, 1]$: The discount factor that weighs future rewards relative to immediate rewards.

As shown in Fig. 8, agent $u \in \mathcal{U}$ selects an action a_u^t based on the current state s^t at each time step t . The system then transitions to a new state s^{t+1} and agent u receives a reward $R_u(s^t, a^t, s^{t+1})$. Each agent aims to maximize its cumulative reward over time by determining a policy $\pi_u : \mathcal{S} \rightarrow \Delta(\mathcal{A}_u)$, where action a_u^t is chosen according to $\pi_u(\cdot | s^t)$. The value function $V_u : \mathcal{S} \rightarrow \mathbb{R}$ for agent u depends on the joint policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$, defined as $\pi(a | s) := \prod_{u \in \mathcal{U}} \pi_u(a_u | s)$. For a given joint policy π and state $s \in \mathcal{S}$, the value function $V_{\pi_u, \pi_{-u}}^u(s)$ is defined as

$$\mathbb{E} \left[\sum_{t \geq 0} \gamma^t R_u(s^t, a^t, s^{t+1}) \mid a_u^t \sim \pi_u(\cdot | s^t), s^0 = s \right]. \quad (27)$$

Here, $-u$ refers to all agents except agent u . In MGs, the optimal policy for each agent is influenced by the policies of the other agents. Each agent aims to find an optimal policy to maximize cumulative reward, with Nash equilibrium (NE)

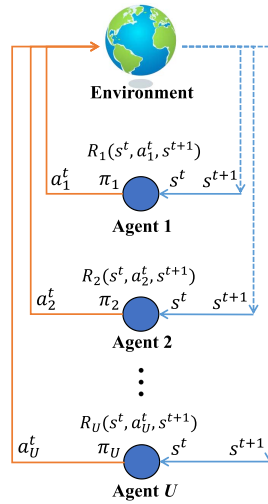


Fig. 8. The architecture of MARL in AASs.

being a common solution concept in MGs, representing a state where no agent can improve its reward by unilaterally changing its policy. In MARL, training involves optimizing policies using acquired experience, including states, actions, and rewards. Execution, in contrast, refers to agents interacting with the environment by taking actions based on their individual or joint policies. Depending on whether agents require information from others during training and whether external information is needed during execution, MARL architectures can be categorized into four types: fully decentralized learning, fully centralized learning, centralized training with decentralized execution (CTDE), and decentralized learning with networked agents [48], [49].

- *Fully Decentralized Learning.* This approach extends single-agent RL to multi-agent environments, where each agent independently optimizes its policy based only on local observations. With no inter-agent communication, this method is suitable for environments with extremely limited or poor communication conditions. However, it struggles with non-stationarity issues and lacks guarantees for policy convergence.
- *Fully Centralized Learning.* In fully centralized learning, a central controller gathers information from all agents, including actions, rewards, and observations, to develop a global policy. While this approach offers better convergence, it incurs significant communication overhead and delays, making it impractical for real-time, large-scale AASs, as information may become outdated in dynamic environments.
- *CTDE.* In this method, during the training phase, a central agent collects data from all agents to learn individual policies. These policies are then distributed to the agents, who use only their local observations to make decisions during execution. This method addresses partial observability and non-stationarity issues during training while enabling efficient decentralized execution, making it the most popular scheme in recent years.
- *Decentralized Learning with Networked Agents.* This method allows agents to exchange information with their

neighbors. By sharing states, actions, and policies, agents can collaborate locally, using both their own and their neighbors' information to make decisions. This reduces computational complexity and enhances scalability, making it ideal for future Internet applications where network entities naturally communicate and collaborate.

In fact, many existing MARL algorithms can flexibly accommodate the aforementioned architectures with the necessary adjustments.

B. Use Cases

This section presents several classical use cases of MARL in AASs, which are summarized in Table VII.

1) *MARL-Enabled Communication:* MARL has gained significant attention in optimizing communication strategies and networking protocols in AASs, enhancing swarm adaptability and robustness in dynamic environments. Specifically, designing effective routing mechanisms in dynamic AASs is challenging due to mobility and limited transmission range. Conventional flooding or greedy search algorithms either incur high communication overhead or exhibit low routing efficiency. To address these issues, the work in [181] proposes an adaptive communication-based routing algorithm for UAV swarms. Their approach integrates a multi-layer perceptron (MLP) determining necessary routing flooding and a GRU for compressing communication data volume. Additionally, an MARL algorithm is employed to enhance cooperative routing decisions. Moreover, the authors in [182] tackle the channel access challenge for cognitive radio-enabled UAV swarms using an MARL approach. They address the uncertainty of primary user activities and the lack of coordination by framing the channel allocation as a one-shot optimization in an MG setting. The solution involves a hybrid RL algorithm that employs an upper confidence bound-Hoeffding strategy integrated with a double deep Q-network (DDQN), effectively overcoming the curse of dimensionality. Furthermore, the authors in [183] study the optimization of relay selection and power allocation for UAV swarms facing jamming threats using MARL. They develop a scheme that leverages network topology, channel states, and shared observations to enhance policy exploration and stabilize learning. Additionally, the use of TL accelerates initial learning and policy optimization. Considering the impact of agent trajectory control on communication within UAV swarms, the authors in [184] propose a joint trajectory control, frequency allocation, and packet routing algorithm to maximize link utility by considering link stability, signal-to-interference-plus-noise ratio (SINR), queuing delay, and UAV residual energy. This algorithm utilizes adaptive multi-agent deep deterministic policy gradient (MADDPG) and swarming behavior to achieve optimal solutions. Each UAV employs an actor network with an LSTM layer for dynamic topology management and a multi-head attentional critic network for policy adjustment through neighbor collaboration.

2) *MARL-Enabled Navigation:* Coordinating multiple autonomous agents for safe navigation in complex, unknown environments is a significant challenge. Traditional shortest path first algorithms, while effective in static networks,

TABLE VII
MARL-ENABLED APPLICATIONS IN AASS

Category	Work	Object	Research Content
MARL-Enabled Communication	[181]	UAV	Adaptive communication-based routing algorithm using MARL.
	[182]	UAV	MARL-based channel access strategy for cognitive radio.
	[183]	UAV	MARL for relay selection and power allocation in jamming conditions, enhanced by TL.
	[184]	UAV	Joint trajectory control, frequency allocation, and packet routing using MADDPG.
MARL-Enabled Navigation	[185]	UAV	MARL-based network-aware model for dynamic navigation.
	[186]	USV	MADDPG-based approach for dynamic navigation and area assignment strategy.
	[187]	UAV	MARDPG and LSTM-based navigation in 3D environments.
	[188]	UAV	Integration of spatial-temporal transformer and MARL for collision-free navigation.
MARL-Enabled Search	[189]	UUV	MARL aided-cooperative search in obstacle-rich underwater environments.
	[190]	UAV/USV/UUV	MARL-based underwater search strategy for heterogeneous AAS.
	[191]	UAV	DNQMIX-based cooperative search with a digital twin driven training framework.
MARL-Enabled Formation Control	[192]	AAS	MARL-based formation control and obstacle avoidance with local decision-making.
	[193]	AAS	Integration of MADDPG with mean field theory for large-scale formation control.
	[194]	UUV	MARL-based generative adversarial IL for formation control using expert trajectories.
	[195]	AAS	Model-free MARL with a Hausdorff distance-based reward formation control.
MARL-Enabled Task Scheduling	[196]	AAS	MARL-based proposer-responder mechanism for task scheduling in dynamic environments.
	[197]	AAS	Deep transfer MARL for task scheduling.
	[198]	UGV	MARL algorithms with information potential field-based reward shaping for task scheduling.
	[199]	UUV	DMAPPO-based task allocation to balance timeliness and energy efficiency.
MARL-Enabled Target Hunting	[200]	UAV	End-to-end cooperative MARL for target tracking, incorporating energy-saving strategies and spatial information entropy for enhanced efficiency and coverage.
	[201]	UAV	MARL algorithm integrating expert knowledge with joint state abstraction via dynamic IDs and Q-value initialization using an artificial potential field for accelerating convergence.
	[202]	UAV	Modified MADDPG method assigning heterogeneous roles with Voronoi-based reward to balance exploration and exploitation for dynamic target tracking.
	[203]	UAV	MADDPG-based strategy with a robust tracking controller and energy-efficient compensator.
	[204]	AAS	Scalable MARL for target hunting, employing bidirectional RNNs in policy and Q networks, enabling adaptability to agent failures or additions with shared parameters.
MARL-Enabled Confrontation	[205]	UAV	MADDPG-based approach for managing coordination and strategy in confrontations.
	[206]	AAS	MADDPG-aided sea battlefield confrontations, incorporating LSTM and actor-critic methods to handle large state spaces and sparse rewards, with IL for faster convergence.
	[207]	UAV	Hierarchical attention actor-critic aided MARL approach for confrontations.
	[208]	UAV	Graph attention aided MARL for confrontations.
	[209]	UAV	Bio-inspired MARL for attack-defense confrontations, based on grouping mechanisms and bio-inspired action space.
	[210]	AAS	Consensus action policy enhanced MARL by promoting cooperative joint actions to reduce team failures in confrontations.

become suboptimal under dynamic conditions and can cause congestion by routing multiple vehicles along the same path. To address this, the authors in [185] propose a network-aware MARL model, where each agent acts as a router. These agents provide routing instructions based on their destinations, the current state of the road network, and learned policies from cooperation with neighboring agents. Furthermore, the authors in [186] present an MADDPG-based cooperative approach to dynamic navigation and area assignment strategy for USV swarms, focusing on optimizing USVs' trajectories, dynamic obstacle avoidance, and coordination. The authors in [187] develop a novel multi-agent recurrent deterministic policy gradient (MARDPG) algorithm to enhance multi-UAV navigation in complex and dynamically changing 3D environments, which uniquely integrates LSTM networks to utilize historical data, thereby improving action-value predictions and overcoming environmental uncertainties. By centralizing learning in each critic network, MARDPG enables UAVs to predict and align with each other's policies, addressing the common issue of slow convergence in decentralized systems. Additionally, parameter sharing among

critic networks facilitates faster training without the need for inter-UAV communication. To ensure collision-free navigation for AASs in human-shared environments while adhering to social norms, despite limited inter-robot communication, the work in [188] introduces a novel simulation environment. This environment is based on decentralized partially observable Markov decision processes (Dec-POSMDP) and MARL to narrow the simulation-to-reality gap commonly found in AAS navigation systems. Additionally, they integrate a spatial-temporal transformer with MARL. This method significantly improves the autonomous agents' ability to interpret and react to human-agent interactions, fostering cooperative behavior among autonomous agents in socially dynamic settings.

3) *MARL-Enabled Search*: Finding sparse targets in an unknown environment by proactively making data-collection decisions is an important task. Current algorithms, while theoretically sound, exhibit limitations: they are often myopic regarding information gain, introduce biases that hinder truly non-myopic behavior, use general-purpose coordination mechanisms unsuitable for specific agent configurations, and require significant online computation for selecting sensing regions.

The authors in [189] propose employing MARL for UUV swarm cooperative search. They model the search task as a decentralized POMDP and establish a search information map. Each UUV updates this map using sonar data and information fusion, enabling real-time decisions based on local observations. This approach improves coordination and search efficiency in unknown, obstacle-rich underwater environments by leveraging MARL strategies to better handle limited detection capabilities and observational information. Searching for marine targets involves complexities due to large areas, distinct signal behaviors, and visibility constraints, making it difficult for single-agent or uniform multi-agent systems. Therefore, the authors in [190] present an MARL-based hierarchical heterogeneous multi-agent approach to enhance underwater search missions, integrating UAVs, USVs and UUVs, leveraging their varied operational capabilities to improve performance over traditional systems. Regarding the UAV swarm cooperative target search issue, the authors in [191] propose DNQMIX, which improves target location, area coverage, and threat avoidance by using a novel reward function for efficient exploration and exploitation. Additionally, a digital twin-driven framework—"centralized training, decentralized execution, and continuous evolution"—is designed to balance training speed and fidelity.

4) *MARL-Enabled Formation Control*: The formation control problem, a fundamental issue in AAS collaboration, involves guiding multiple agents to form a specified arrangement. Traditional methods, such as model predictive control and fuzzy control, rely on precise global information and are often inadequate in harsh environments. Additionally, RL-based approaches using a leader-follower structure can compromise agent collaboration, leading to maneuverability and robustness issues. To address these challenges, the authors in [192] present an MARL-based solution with distributed formation and obstacle avoidance. This approach allows agents to make decisions and control themselves using local and relative information, enabling quick reorganization into new topologies if any agent becomes disconnected. To further enhance the effectiveness of large-scale multi-agent formations, the authors in [193] integrate mean field theory into the MADDPG algorithm to address the formation control issue. Moreover, in complex underwater environments, the authors in [194] implement multi-agent generative adversarial IL for formation control and obstacle avoidance using expert-demonstrated trajectories, effectively addressing the difficulty of designing reward functions. They employ a decentralized training with decentralized execution framework to mitigate underwater communication issues. Additionally, the accuracy and convergence speed of the discriminator are improved by dividing expert state-action pairs into two groups and updating it with randomly selected pairs. Furthermore, to improve robustness in formation control, the authors in [195] tackle the challenge of developing robust distributed control policies for flexible formation control without relying on a global coordinate system. They propose a fully decentralized displacement-based formation control policy using a model-free MARL approach, trained centrally, where the Hausdorff distance is incorporated into the reward function to measure the difference between the current and target topologies.

5) *MARL-Enabled Task Scheduling*: In dynamic and uncertain environments, task scheduling for AASs is crucial for effective operations. Conventional task scheduling methods often rely on centralized decision-making, resulting in single points of failure, poor scalability, and difficulty adapting to dynamic environments. MARL addresses these limitations by enabling decentralized collaboration and joint strategy optimization among agents, making it more effective for task scheduling in AASs. The authors in [196] introduce a proposer-responder mechanism and prioritized experience replay within an MARL framework. By establishing a coordinated network to manage task proposals and using a Q-network to predict task participation outcomes, this method effectively handles dynamic task uncertainties and enhances online computation efficiency. The authors in [197] propose to employ deep transfer MARL to enhance decision-making efficiency of task scheduling. This method involves comparing new tasks with previously encountered ones to identify similarities, followed by transferring pre-trained network parameters to expedite the learning process and improve algorithm convergence. For UGV swarms, the authors in [198] propose two types of MARL algorithms, which utilize an information potential field-based reward shaping strategy. This strategy provides stepwise rewards to guide UGVs towards different targets, addressing the challenge of reward sparsity and enhancing distribution efficiency. In the context of underwater multi-objective tasks, considering timeliness and energy consumption, the authors in [199] present an age of information (AoI) based cooperative searching approach and an energy-efficient distributed multi-agent proximal policy optimization (DMAPPO) scheme, which autonomously adjusts the number of UUVs per task based on real-time requirements and residual energy, improving both task completion and energy efficiency.

6) *MARL-Enabled Target Hunting*: Target hunting, also referred to as pursuit-evasion, involves capturing mobile targets using one or more pursuers. While it shares similarities with search tasks, it is inherently more complex and can be regarded as encompassing search within its scope. The quantitative definition of a successful encirclement typically involves multiple agents achieving a state of control over the target, characterized by specific criteria such as maintaining a certain distance from the target or forming an enclosure with a predefined angular configuration. Traditional methods rely on preset trajectories, which are impractical due to unpredictable target movements. Therefore, the authors in [200] propose an end-to-end cooperative MARL scheme to enhance UAV target tracking, where the agents make their decisions based on target states. Additionally, they introduce energy-saving strategies to reduce power usage and extend system operation time, and spatial information entropy to improve detection coverage. Furthermore, an MARL algorithm leveraging expert knowledge is presented in [201]. First, they introduce a multi-agent joint state abstraction method based on dynamic IDs to reduce the state space significantly. Second, they employ a multi-agent Q-learning algorithm utilizing an artificial potential field to initialize Q-values using environmental knowledge, thus providing a solid foundation for learning and speeding up convergence. Similarly, the authors

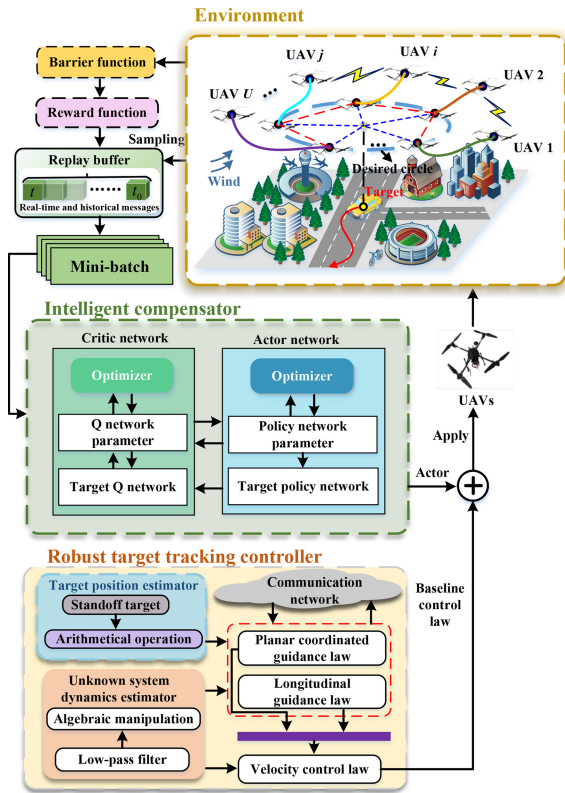


Fig. 9. A scenario of MARL-enabled target hunting.

in [202] propose a modified MADDPG method which assigns heterogeneous roles to UAVs for balancing exploration and exploitation, enabling effective target tracking and discovery in dynamic environments with random obstacles. Additionally, the proposed algorithm employs a Voronoi-based rewarding policy to further facilitate exploration. Taking strict time constraints and prolonged target tracking requirements into account, the authors in [203] develop an MADDPG-based target hunting strategy for networked UAV swarm to address time constraints while reducing energy expenditure. This strategy consists of a robust target tracking controller for driving UAVs to encircle an escape target with a desired formation pattern despite wind disturbances and an intelligent compensator for the optimal index in terms of tracking accuracy and energy consumption by devising barrier functions. Fig. 9 illustrate an example of MADDPG-based target hunting strategy for UAV swarm. In response to scalability challenges where traditional schemes fail during agent failures or when new entities are introduced, a novel scalable MARL approach for target hunting is introduced in [204]. This method employs a bidirectional RNN within the policy and Q networks to address scalability issues effectively. The shared parameters in the bidirectional RNN structure ensure that the system can adapt when agents are added or removed, utilizing existing parameters to maintain functionality.

7) *MARL-Enabled Confrontation*: Attack-defense confrontation is an advanced form of the defending territory game, where a swarm of intruder agents attempts to invade a territory guarded by defender agents. This scenario significantly challenges traditional methods. To address this, the authors

in [205] present an MADDPG-based confrontation approach that effectively manages the coordination and strategic planning required in such dynamic and adversarial settings. Furthermore, the authors in [206] extend MADDPG to sea battlefield confrontations, combining it with LSTM and actor-critic methods to ensure convergence in large state and action spaces while addressing the issue of sparse rewards. In [207], the authors introduce the hierarchical attention actor-critic algorithm to address decision-making and coordination in large-scale UAV swarm confrontations. This approach employs MARL and integrates a centralized critic network with a hierarchical two-stage attention network, which dynamically captures interactions among UAVs. Similarly, the authors in [208] introduce the graph attention mechanism and soft actor critic (SAC) into MARL with target predicting network, which incorporates trajectory prediction with MARL for enhanced decision-making. By adapting the SAC for multi-agent settings and incorporating a graph attention neural network, this method optimizes collaboration by focusing on spatial interactions among UAVs. Additionally, IL is incorporated to accelerate convergence and enhance the algorithm's effectiveness. Considering the exponential increase in training time as swarm size grows and inspired by natural group hunting behaviors, the authors in [209] introduce a bio-inspired decision-making method for UAV swarm attack-defense confrontations. They establish a UAV swarm decision-making framework based on grouping mechanisms, complemented by a bio-inspired action space. Addressing the inability of MARL to prevent failure-inducing actions, the authors in [210] highlight the shortcomings of existing joint action policies in MARL for AAS confrontations. They propose a consensus action policy that records the frequency of joint actions leading to failures and integrates a cooperation tendency with each agent's Q-value and Nash bargaining solution. This approach encourages cooperative actions, avoids those prone to failure, and is adaptable to partially observable scenarios using DQN or actor-critic methods.

C. Key Techniques

1) *Learning Framework*: The foundational framework of MARL consists of three key elements: learning strategies [211], [212], [213], [214], [215], [216], agent modeling [217], [218], [219], [220], [221], [222], and credit assignment [223], [224], [225], [226], [227], [228]. Table VIII summarizes the key techniques of MARL applied in AASs.

- *Learning Strategies*. Learning strategies guide how agents adapt and optimize their decision-making processes in dynamic and uncertain environments, enhancing task performance through continuous improvement. These strategies fall into three categories: policy gradient-based [211], [212], value decomposition-based [213], [214], and integration of both approaches [215], [216]. Policy gradient-based strategies, such as the MADDPG algorithm introduced in [211], extend the traditional DDPG to MARL. This method employs centralized training with distributed execution, utilizing actor and critic networks. The critic

TABLE VIII
SUMMARY OF KEY TECHNIQUES OF MARL APPLIED IN AASS

Category	Subcategory	Approaches or key methods
Learning framework	Learning strategies	MADDPG [211]; COMA [212]; VDN [213]; QMIX [214]; VDAC [215]; DOP [216]
	Agent modeling	VAE-based opponent modeling [217]; Latent strategy learning [218]; MBOM [219]; TDOM [220]; Transformer-based trajectory prediction [221]; CLAM [222]
	Credit assignment	MLCA [223]; MAPPG [224]; Shapley counterfactual credit assignment [225]; AIRMN [226]; STAS [227]; MACD [228]
Knowledge reuse	Parameter sharing	Composite DNN architecture [230]; Automatic role assignment [231]; Diversity regularization [232]; Brain-inspired sharing [233]; Role-based sharing [234]; FL-inspired sharing [235]
	TL-based solutions	MKT-MARL [236]; MAPTF [237]; Parallel TL [238]; MALT [239]; Meta-knowledge transfer [240]
	Curriculum learning-based solutions	Evolutionary population curriculum [241]; VACL [242]; Dynamic agent-number network [243]; Subgame curriculum learning [244]
Sample utilization	Exploration	MAVEN [246]; CMAE [247]; EMC [248]; UneVEn [249]; COIN [250]
	Model-based solutions	MAMBA [251]; Guided diffusion [252]; Implicit model-based MARL [253]; MAG framework [254]
Task simplification	Hierarchical learning	HAVEN [255]; HSTCN [256]; HiSOMA [257]; HCTA [258]; Intrinsic reward rectification [259]
	Topological structure learning	Graph convolution [260]; DAG-based coordination [261]; Graph attention mean field [262]; DICG [263]
Communication strategy	Preset communication	DIAL [264]; CommNet [265]; TMC [266]; Discrete token communication [267]; IS [268]; MAIC [269]
	Learnable communication	TarMAC [270]; IC3Net [271]; LSC [272]; Graph information bottleneck optimization [273]; HetNet [274]

accesses global information during training to update the actor, which uses local information during testing. MADDPG also features additional networks to predict strategies of other agents, achieving success in various tasks. In contrast, the study in [212] presents the counterfactual multi-agent policy gradients (COMA) algorithm, which enhances collaboration by using a global critic to assess all actions and states. It computes rewards by contrasting the overall reward resulting from a changed agent action with the current reward. Value decomposition-based strategies like the value decomposition network (VDN) proposed in [213] address the “lazy agent” phenomenon by decomposing the global value function into individual functions, simplifying the action space. QMIX, introduced in [214], employs a mixing network that ensures consistency between joint and individual action-values, enhancing coordination by maintaining a monotonic relationship through non-negative weights. Hybrid strategies combine the strengths of the previous approaches for balanced efficiency and performance. For instance, the work in [215] introduces the value-decomposition actor-critic (VDAC) strategy, extending value decomposition to actor-critic methods, while the work in [216] proposes decomposed policy gradient method (DOP), which integrates value decomposition within a multi-agent actor-critic framework. This setup divides the centralized critic into individual critics weighted by local actions, facilitating scalable learning and offering advantages like feasible off-policy evaluation for stochastic policies and mitigating inconsistencies between centralized training and decentralized execution.

- *Agent Modeling.* Agent modeling aims to predict and understand the behaviors of other agents, enhancing coordination and competition in multi-agent systems [179], [229]. In [217], a method is introduced for opponent modeling using variational autoencoders (VAE) that controls an agent within a multi-agent system,

facilitating successful interactions with strategically fixed opponents. The proposal includes two VAE-based methods: opponent modeling DDPG (OMDDPG), which utilizes opponent information, and self-modeling A2C (SMA2C), which does not. Moreover, the work in [218] presents a novel framework for multi-agent interaction that concurrently learns latent strategies and responsive policies. This framework implicitly models the evolution of another agent’s strategy in response to the ego agent’s actions. Through repeated interactions, the ego agent abstracts each opponent’s policy into a high-level latent strategy and learns that opponents adapt to avoid areas recently blocked by the ego agent. Utilizing these dynamics, the ego agent strategically develops a policy to maneuver opponents into positions where blocking is most effective. In [219], model-based opponent modeling (MBOM) is proposed, leveraging an environmental model to adapt to various opponents and simulate recursive reasoning processes, thereby generating diverse, improving strategies. It also blends these strategies based on their similarity to actual opponent behaviors, enhancing representation accuracy and effectiveness. In [220], a time dynamical opponent model (TDOM) is introduced that encodes the tendency of opponent policies to evolve over time. The model’s theoretical foundation is established by deriving a lower bound for the log objective of individual agents. In [221], an innovative opponent modeling method is introduced, focusing on the episodic trajectories of other agents. This approach transforms opponent modeling into a sequence modeling challenge, employing a transformer model to forecast opposing agents’ trajectories using the agent’s local trajectory sequence. Given that many existing agent modeling methods rely on local observations or extensive observation trajectories, limiting applicability, the work in [222] proposes a contrastive learning-based method (CLAM) that uses only the ego agent’s local observations during training and execution.

- *Credit Assignment.* Credit assignment assesses each agent's contribution to swarm performance, especially when collective rewards cannot be directly linked to individual actions, aligning individual incentives with group objectives. In [223], a multi-level credit assignment (MLCA) approach is introduced, encompassing both manager and worker levels. At the manager level, a dilated GRU manages high-level plans, whereas at the worker level, a GRU executes primitive actions based on these plans. A centralized critic is assigned to each level to learn its specific credit assignment scheme. In [224], multi-agent polarization policy gradient (MAPPG) is proposed, utilizing a polarization function to align joint and individual actions into manageable constraints, thus enhancing credit assignment efficiency. Theoretically, it is demonstrated that MAPPG's individual policies can converge to the global optimum. Given that many existing credit assignment methods inadequately address complex agent interactions by focusing solely on decomposing the joint value function, the study in [225] introduces Shapley counterfactual credit assignment. This novel approach utilizes Shapley values to account for agent coalitions and accurately estimate individual credits. Traditional methods typically combine intrinsic with global rewards to form a total reward; however, they often overlook agent interdependence, reducing their adaptability and effectiveness. In response, the authors in [226] propose an attention-based intrinsic reward mixing network (AIRMN), addressing these limitations. AIRMN utilizes an attention mechanism to enhance teamwork efficacy and features a dynamic mixing network that nonlinearly combines intrinsic and extrinsic rewards, adapting to environmental changes. Addressing inefficiencies in modeling delayed global rewards, the study presented in [227] introduces spatial-temporal attention with Shapley (STAS). This method decomposes global returns at each timestep, using Shapley values to redistribute individual payoffs. Furthermore, the work in [228] introduces multi-agent counterfactual dreamer (MACD), which tackles credit assignment and non-stationarity through counterfactual trajectories based on a learned world model, optimizing the multi-agent learning objective.

2) *Knowledge Reuse:* Knowledge reuse can significantly accelerate learning speed and enhance performance even with limited samples, effectively addressing the scalability challenges of MARL. Widely used methods include parameter sharing [230], [231], [232], [233], [234], [235], TL-based solutions [236], [237], [238], [239], [240], and curriculum learning-based solutions [241], [242], [243], [244].

- *Parameter Sharing.* Parameter sharing accelerates policy development by facilitating knowledge transfer among agents, thereby improving learning efficiency and reducing the overall number of required parameters, which significantly shortens training times. The authors in [230] employ a composite DNN architecture to compute the factorized Q-function, emphasizing parameter sharing among all agents within the same group. However, universal parameter sharing can hinder learning by causing

all agents to develop similar strategies, which may not be optimal for complex tasks. In response, the work presented in [231] highlights the potential drawbacks of indiscriminate parameter sharing, noting that its effectiveness depends heavily on the environment. They introduce a method that automatically identifies agents who may benefit from parameter sharing by categorizing them based on their abilities and goals, blending the sample efficiency of parameter sharing with the representational capacity of independent networks. Conventional parameter sharing methods may cause agents to behave similarly, limiting their coordination capacity. To address this, the work in [232] introduces diversity in optimization and representation within parameter sharing. They implement an information-theoretic regularization to maximize the mutual information between agents' identities and their trajectories, encouraging diverse behaviors. Additionally, agent-specific modules within the shared network architecture are regularized by L1-norm to maintain diversity while promoting shared learning. Simple parameter sharing often fails to accommodate diverse strategies among agents with varying identities or tasks. To overcome this, the study in [233] introduces a parameter sharing technique inspired by biological brain research, assigning each type of agent to specific regions within a shared network to create distinct subnetworks. This approach enhances strategy diversity among agents without increasing the number of training parameters. The authors in [234] propose a role parameter sharing scheme, dynamically dividing roles and selecting entities for parameter sharing. This approach identifies agents that could benefit from sharing and implements partial parameter sharing within these roles. Additionally, intrinsic rewards are introduced to enhance strategy diversity. Inspired by FL, the work presented in [235] proposes three innovative parameter sharing approaches: average periodically parameter sharing, reward-scalability periodically parameter sharing, and partial personalized periodically parameter sharing. These methods involve agents periodically sharing their Q-value networks, adapting collected rewards as scalability factors, and updating parts of their neural networks during the sharing period.

- *TL-Based Solutions.* TL can expedite the MARL learning process by allowing agents to apply previously acquired knowledge to new and similar tasks [245]. In [236], a novel multi-task method based on knowledge transfer in cooperative MARL (MKT-MARL) is proposed. Utilizing task-specific instructors, this method enables a team of agents to achieve expert performance across various tasks. MKT-MARL employs a KD algorithm designed for multi-agent systems, incorporating teacher insights and using a teacher annealing strategy to gradually transition from KD to focusing on environmental rewards, thereby boosting performance. A multi-agent policy transfer framework (MAPTF), which enhances MARL efficiency through inter-agent knowledge transfer, is presented in [237]. MAPTF views policy transfer among agents as an option learning problem, enabling adaptive

policy selection for agents and timely termination to avert adverse effects. The framework addresses sample inconsistency by introducing successor representation option learning, which separates environmental dynamics from rewards, allowing precise learning of option-value functions suited to individual agent preferences. In [238], parallel TL is proposed to enable simultaneous learning and knowledge transfer between tasks, enhancing online performance by introducing methods for selecting transferable knowledge, determining transfer frequency and size, and integrating knowledge into the target task. The authors in [239] propose multi-agent lateral transfer (MALT), a versatile cross-task TL method aimed at reducing the training burden in MARL. This method tackles challenges in developing a universal cross-task transfer method by reusing features instead of policies or experiences, drawing inspiration from progressive networks. MALT assigns clustered agents pre-trained policy networks and integrates an attention module to improve the transfer framework. It supports knowledge transfer among diverse agents and prevents negative transfer, even in completely distinct tasks, unlike existing methods. Additionally, the work in [240] introduces the transfer of meta-knowledge to improve exploration through coordinated schemes by reusing meta-experience from auxiliary tasks. This utilizes a weighting function within the monotonic loss function to prioritize important joint actions and avoid suboptimal policies, enhancing the effectiveness of MARL strategies.

- *Curriculum Learning-Based Solutions.* Curriculum learning in MARL is inspired by human educational processes, systematically building agent capabilities through sequenced stages of development tailored to the agents' current skills. The evolutionary population curriculum, proposed in [241], incrementally increases the population size and employs evolutionary strategies to align objectives more effectively. This method enhances performance by selectively refining and integrating the best-performing agents at each stage, ensuring they adapt well as the population expands. Furthermore, the work in [242] introduces variational automatic curriculum learning (VACL), a goal-conditioned cooperative MARL algorithm that employs a variational approach to split the learning objective between task learning and curriculum updating. It methodically expands training tasks from simpler to more complex, effectively addressing sparse-reward challenges involving multiple agents. In another approach, the work in [243] details dynamic multi-agent curriculum learning that starts with a small number of agents and systematically increases this number. This method enhances learning efficiency through three transfer mechanisms and introduces the dynamic agent-number network (DyAN), which adapts to varying state dimensions across different curricula, addressing the limitations of fixed-size network inputs. Additionally, the authors in [244] propose a subgame curriculum learning framework that uses an adaptive initial state distribution, resetting agents to previously encountered

states where they can quickly improve performance. This framework also incorporates a subgame selection metric that approximates the squared distance to NE values and employs a particle-based state sampler to generate challenging subgames.

- 3) *Sample Utilization.* The primary methods to improve sample utilization include exploration [246], [247], [248], [249], [250] and model-based solutions [251], [252], [253], [254].

- *Exploration.* Exploration significantly enhances sample utilization efficiency, especially in tasks with sparse rewards. In such frameworks, agents learn optimal behaviors by experimenting with new strategies and actions to uncover unknown information. This helps them break away from known but potentially suboptimal strategies in favor of more effective solutions through unexplored actions. Addressing the limitations of joint action-values representation in QMIX, which leads to poor exploration and performance, the authors in [246] introduce the multi-agent variational exploration (MAVEN) method. This approach combines value-based and policy-based strategies through a latent space for hierarchical control, enabling agents to condition their actions on a shared latent variable controlled by a hierarchical policy. This setup fosters committed and extended exploration necessary for complex multi-agent environments. Recognizing the challenge where agents fail to identify valuable states for exploration and struggle with coordination, [247] proposes the cooperative multi-agent exploration (CMAE) method. In this method, agents share a common exploration goal selected via a normalized entropy-based method from multiple projected state spaces, promoting coordinated exploration. In another approach, the work in [248] presents a curiosity-driven exploration (EMC) strategy that uses individual Q-values as embeddings of local action-observation histories to capture agent interactions. This method encourages exploration through reward backpropagation during centralized training and employs prediction errors of these Q-values as intrinsic rewards. Episodic memory is also utilized to capitalize on informative experiences, further enhancing policy training. Moreover, the work in [249] introduces the universal value exploration (UneVEN) strategy, which improves joint exploration by learning a set of related tasks through the linear decomposition of universal successor features. This strategy not only boosts agent coordination but also draws on policies from previously solved tasks to enhance joint exploration efforts. Finally, recognizing the computational demands of exploring unknown states or measuring agent interactions, the authors in [250] propose the curiosity-based and influence-based exploration (COIN) approach. COIN evaluates each agent's impact on others using mutual information theory, applying this as intrinsic rewards to individual value functions and combining it with curiosity-driven rewards from prediction errors, fostering comprehensive and effective exploration in cooperative MARL tasks.

- *Model-Based Solutions.* Although most MARL algorithms operate on model-free principles, relying heavily on continuous interaction with the environment for policy optimization, they tend to suffer from low sample efficiency. In contrast, model-based MARL enhances sample efficiency by learning the state transition and reward functions of the environment, referred to as the world model. This allows for significant enhancement of policy optimization through the knowledge derived from the world model, resulting in improved sample efficiency. The authors in [251] present the multi-agent model-based (MAMBA) approach, which boosts centralized training in cooperative environments. This method enables each agent to maintain a world model during execution through agent communication and uses imaginary rollouts for training, thus reducing the need for continuous environmental interaction. However, as the number of agents increases, most model-based MARL schemes struggle with world model construction due to the accumulation of redundant information. To address this, the work in [252] introduces a method that leverages guided diffusion to filter out noisy data, disrupting teamwork. This cleaned data helps construct a unified world model that denoises individual agent observations and aids in planning for multi-agent policy optimization, thereby enhancing cooperative efficiency. The complexity inherent in multi-agent systems complicates learning environment models, with compounded errors potentially hindering progress. In response, [253] introduces an implicit model-based MARL method grounded in value decomposition. This approach enables agents to engage with a learned virtual environment and evaluate current state values by envisioning future states within a latent space, thereby providing valuable foresight into potential outcomes. Despite advancements in model-based reinforcement learning, the complexity due to the exponential growth of the joint state-action space in multi-agent systems presents significant challenges. This complexity often necessitates the use of agent-wise local models rather than a comprehensive global model. However, predictions from local models in multi-step rollouts can influence each other, leading to compounded prediction errors and global inaccuracies. To mitigate this, [254] introduces the models as agents (MAG) framework, which treats local models as multi-step decision-making agents and considers current policies as dynamics during rollouts. This innovative approach allows local models to account for their mutual effects over multiple steps, aiming to enhance prediction accuracy significantly.
- 4) *Task Simplification:* In complex scenarios, MARL may face the challenge of the curse of dimensionality. Two categories of methods are available to alleviate this issue: hierarchical learning [255], [256], [257], [258], [259] and topological structure learning [260], [261], [262], [263]. These approaches effectively address the complexities inherent in the state and action spaces, thereby enhancing the overall efficiency of the learning process.
- *Hierarchical Learning.* Hierarchical MARL (HMARL) integrates hierarchical structures with AASs to improve decision-making in complex environments by decomposing intricate problems into manageable sub-tasks, thereby reducing computational complexity. The authors in [255] introduce the hierarchical value decomposition (HAVEN) framework for fully cooperative multi-agent problems, which is inspired by the coordination mechanisms of the human nervous system. It employs a dual coordination strategy with two-level hierarchical reward functions to handle instability from simultaneous policy optimization across different levels and agents, requiring no prior domain knowledge or pre-training and being adaptable to any value decomposition variant. Furthermore, the hierarchical spatio-temporal communication network (HSTCN) detailed in [256] utilizes a dual-level policy framework. The high-level policy generates intrinsic goals and continuous reward samples to mitigate reward sparsity, while the low-level policy uses this data to refine agent execution and interaction with the environment. Moreover, the high-level policy employs a graph-based model to capture spatial and temporal dependencies, enhancing the learning of robust policies. In [257], a hierarchical multi-agent model integrating self-organizing neural networks with MADRL (HiSOMA) is proposed for long-horizon, multi-task decision-making. HiSOMA's top-level controller, fusion architecture for learning, cognition, and navigation (FALCON), uses self-organizing neural networks to learn high-level decision rules that influence middle-level controllers. These middle-level controllers manage bottom-level controllers executing individual action policies and interacting directly with the environment. The hierarchical cooperative task allocation (HCTA) framework, proposed in [258], tackles dynamic task allocation in fully cooperative multi-agent systems. It trains a task selector based on action chains to improve efficiency and introduces a hierarchical cooperative mechanism to parallelize upper-level task selection with lower-level task decision-making, thereby accelerating the learning of cooperative strategies. Additionally, addressing the often-overlooked impact of macro actions on decision-making in HMARL, which affects the distribution of intrinsic rewards, the work in [259] introduces a value decomposition-based HMARL method. This method incorporates intrinsic reward rectification to determine the effectiveness of macro actions and correct intrinsic rewards, enhancing system performance.
- *Topological Structure Learning.* Topological structure learning seeks to characterize relationships between agents and reduce the joint action space of AASs in complex scenarios. However, cooperating in dynamic multi-agent environments is challenging due to the constant movement of agents and their changing neighbors, complicating the learning of mutual interplay representations. To address these challenges, the authors in [260] propose a graph convolutional RL approach that adapts graph convolution to the dynamic nature of multi-agent settings and uses relation kernels to

capture agent interplay through relation representations. Latent features from convolutional layers with gradually increasing receptive fields enhance cooperation, complemented by temporal relation regularization to maintain consistency. Moreover, the authors in [261] introduce a novel method by factorizing the joint team policy into a graph generator and a graph-based coordinated policy to foster coordinated behaviors among agents. The graph generator employs an encoder-decoder framework to output DAGs, capturing dynamic decision structures and using DAGness and DAG depth constraints for optimization. The coordinated policy leverages these generated decision structures, with both the graph generator and the coordinated policy simultaneously trained to maximize the discounted return. In another innovative approach, the work in [262] introduces the graph attention mean field, which simplifies complexity by transforming agent-agent interactions into interactions between each agent and a weighted mean field. This method precisely models interaction dynamics and incorporates a graph attention mechanism to capture diverse and time-varying interaction strengths. Fully centralized control in MARL is often impractical due to large joint action spaces. Coordination graphs provide a structured approach to reasoning about joint actions but typically require domain-specific expertise. Addressing this, [263] proposes the deep implicit coordination graph (DICG) architecture, which autonomously infers dynamic coordination graph structures. These structures are utilized by a GNN module to learn implicit reasoning about joint actions or values. DICG balances full centralization and decentralization using standard actor-critic methods, significantly enhancing coordination in environments with numerous agents.

5) *Communication Strategy*: Communication is pivotal in MARL, as it allows agents to exchange critical information such as experiences, intentions, and observations through various communication technologies. Existing communication strategies can be divided into preset communication and learnable communication.

- *Preset Communication*. Preset communication protocols establish fixed methods for agents exchange information. In [264], the authors propose differentiable inter-agent learning (DIAL), where each agent broadcasts messages to every other agent. CommNet, introduced by [265], manages each agent with a deep feed-forward network that accesses a communication channel transmitting a continuous vector. This channel aggregates transmissions from all agents, enabling each agent to learn autonomously what to transmit, rather than adhering to predefined rules. The continuous nature of this communication permits training via back-propagation, making the model compatible with standard single-agent RL algorithms or supervised learning techniques. Addressing the challenge of excessive message exchange, the authors in [266] introduce temporal message control (TMC), which employs a temporal smoothing technique to significantly reduce the communication overhead between

agents, thus enhancing the practicality of MARL in real-world scenarios with unreliable communication channels. In [267], the authors propose an advanced communication method for neural agents, overcoming the limitations of one-hot vector tokens. Inspired by word embedding techniques from natural language processing, this method develops architectures that allow agents to communicate via discrete tokens derived from a continuous space, optimizing communication across various scenarios and enhancing zero-shot understanding. The work in [268] introduces a novel communication scheme called intention sharing (IS), where each agent models the environment dynamics and the actions of other agents to generate an imagined trajectory, representing its future action plan. This trajectory is compressed into an intention message using an attention mechanism that assesses the relative importance of trajectory components based on messages received from other agents. Recognizing that traditional communication schemes involving the sharing of agents' local observations or latent embeddings can increase decision-making uncertainty and complicate learning, leading to suboptimal coordination, the study in [269] proposes the multi-agent incentive communication (MAIC) framework. This framework enables agents to generate incentive messages that directly influence other agents' value functions, promoting explicit coordination. It includes learning targeted teammate models to anticipate actions and create tailored messages, and introduces regularization to leverage interaction sparsity and improve communication efficiency.

- *Learnable Communication*. While preset communication protocols ensure consistency and predictability, they lack flexibility to adapt to new or unforeseen scenarios during agent interactions. Recent interest has grown in learnable communication, which allows agents to dynamically evolve their communication strategies based on environmental and task-specific needs. The authors in [270] introduce targeted multi-agent communication (TarMAC), enabling agents to learn both the content and the recipients of their messages solely from task-specific rewards, without predefined communication rules. This approach is augmented with a multi-round communication strategy, allowing for extensive coordination over several rounds before action. Moreover, the authors in [271] present the individualized controlled continuous communication model (IC3Net), expanding beyond models designed solely for fully cooperative tasks. IC3Net features a gating mechanism for controlled communication and provides individualized rewards for each agent, making it adaptable to both semi-cooperative and competitive environments. The authors in [272] present learning structured communication (LSC), a framework that develops a flexible hierarchical communication topology through two modules: a structured communication module for optimal topology formation and a communication-based policy module utilizing hierarchical graph neural networks for message generation, information propagation, and action selection.

Furthermore, the work in [273] proposes a robust communication learning mechanism using graph information bottleneck optimization. This method enhances robustness and effectiveness in communication by employing two information-theoretic regularizers that maximize the mutual information between message representation and action selection while minimizing it between agent features and message representation. Addressing the limitations of homogeneous graph networks, which often fail to account for agent heterogeneity and thereby reduce communication effectiveness, heterogeneous policy networks (HetNet) is proposed [274]. The method utilizes heterogeneous graph-attention networks to facilitate the learning of collaborative policies tailored to each agent class, enabling end-to-end training for binarized messaging.

Lessons Learned: This section surveys the application of MARL in AASs, covering a wide range of mission scenarios including collaborative communication, navigation, search, formation control, task scheduling, and adversarial coordination. It also systematically reviews key mechanism designs across learning frameworks, knowledge reuse, sample utilization, task simplification, and communication strategies. While most existing approaches adopt the CTDE paradigm, their reliance on centralized critics and synchronous training often becomes problematic in real-world AASs, where agents operate asynchronously, under partial observability, and with unstable communication. The effectiveness of a MARL algorithm is therefore not determined by its structural sophistication alone, but by how well it aligns with mission requirements and system constraints. Agent modeling and credit assignment enhance coordination under complex interactions but are often constrained by computational overhead and sensitivity to reward structures. For knowledge reuse, naive parameter sharing accelerates training but may lead to policy homogenization, whereas role-aware or periodic sharing strategies offer better adaptability for heterogeneous swarms. To improve sample efficiency, exploration-enhanced and model-based methods are promising but introduce additional communication and computational burdens that must be balanced against system capabilities. In high-dimensional state-action spaces, hierarchical and graph-based approaches can simplify policy learning but may introduce unnecessary complexity if not grounded in task semantics or swarm structure. Communication design must also be task-aware: preset protocols suit stable environments, while learnable communication provides flexibility for dynamic tasks—provided the system can support their overhead. In sum, the successful deployment of MARL in AASs hinges not on algorithmic novelty alone, but on the co-adaptation of task demands, system capabilities, and learning mechanisms.

VII. FUTURE RESEARCH DIRECTIONS

A. Distributed Inference for AASs

Besides model training, model inference in AASs also faces significant challenges, such as energy and computational limitations, particularly when large models are employed. The

inference speed may fail to meet the real-time requirements in dynamic environments. Distributed inference, which alleviates the computational burden on individual agents by distributing tasks across multiple nodes, is suitable to AASs. However, compared to ideal environments, AASs present unique characteristics. The system must adapt to the continuous changes in agents and their environments, requiring dynamic adjustments in task coordination. Additionally, communication links in AASs are often unreliable due to the time-varying topology and intermittent connectivity between agents, which necessitates optimization of data transmission to avoid delays impacting real-time decision-making. Furthermore, the heterogeneity of the agents, with varying computational capacities and resource availability, demands that inference tasks be flexibly distributed to leverage the diverse capabilities of the agents. Furthermore, AASs operate in harsh and unpredictable conditions, necessitating robust adaptability and fault tolerance mechanisms to ensure continuous operation despite agent failures or environmental changes.

B. Mixture of Experts Empowered AASs

In real-world applications, agents within AASs are often heterogeneous and possess diverse capabilities, making them well-suited for cooperatively handling complex environments and tasks. The mixture of experts (MoE) [275] approach, which integrates multiple expert models through a gating mechanism to select the most appropriate expert for a given task, is ideally suited to AASs. MoE's strength lies in its ability to harness the expertise of various models, thereby offering more precise and efficient solutions to diverse and intricate tasks. In AASs, MoE can treat each agent as an independent expert model, focusing on specific domains or types of tasks. Through a distributed gating mechanism, the system dynamically selects the most suitable agents for current tasks, enhancing the overall capacity, efficiency, and robustness of AASs.

C. Distributed Generative Artificial Intelligence for AASs

Generative artificial intelligence (GAI) [276], such as GANs, VAEs, and diffusion models, which learn data distributions to generate new data, hold extensive potential for application within AASs. GAI can enhance the autonomous decision-making capabilities of agents by simulating complex environments and generating strategies. For example, generative models can simulate diverse real-world environments, providing abundant training data for the swarm and alleviating data scarcity issues during model training. Within a DML framework, GAI can facilitate the decentralized sharing of generative models among agents, supporting independent or collaborative strategy and prediction generation. This not only accelerates the learning process of the swarm but also improves its adaptability in complex and dynamic environments. Additionally, GAI can assist in hypothesizing scenarios during the exploration of unknown environments, enabling agents to better assess and optimize their strategic actions.

D. Employing DML in Cross-Domain Heterogeneous AASs

Cross-domain heterogeneous AASs provide valuable situational awareness and fast responses to complex tasks. This capability is especially useful in applications like ocean resource exploitation, environmental monitoring, military defense, and emergency response. For instance, they can concurrently collect atmospheric, water quality, and seabed data during marine monitoring, providing a comprehensive ecosystem assessment. However, cross-domain collaborative operations face technical challenges. First, heterogeneity among platforms introduces issues with communication protocols, data fusion, and coordination, especially when integrating DML across domains. Second, DML in cross-domain systems requires precise spatial and temporal synchronization and task coordination, as communication links may be limited or interrupted in dynamic environments, posing robustness challenges for the AASs.

E. Highly Reliable DML for AASs in Adversarial Environments

In adversarial and data poisoning environments, DML in AASs faces various types of threats. These include adversarial attacks—where malicious modifications to model inputs or communication data are used to mislead decision-making processes; data poisoning—involving the injection of false data or contamination of training datasets to degrade learning performance; and model theft and inference attacks—where adversaries exploit model outputs to extract sensitive information or disrupt model functionality. Such threats can compromise intra-swarm collaboration and potentially render missions ineffective. Although there is substantial research on the security of DML, there is a noticeable gap in studies that fully account for the unique characteristics of AASs. One promising direction is the development of adaptive and proactive defense mechanisms that can detect and mitigate adversarial behaviors in real time, providing a dynamic and robust response to evolving threats. Additionally, enhancing collaborative defense strategies within the swarm, where autonomous agents can share security insights and coordinate their responses, will be vital for maintaining swarm integrity and mission success. Another key area is improving data integrity and privacy measures in distributed learning frameworks, ensuring that the system remains resilient without significantly increasing communication burdens. Exploring these avenues will provide valuable insights and methodologies for advancing the security and robustness of AASs operating in hostile and deceptive environments.

VIII. CONCLUSION

This survey provides a comprehensive review of the integration of DML techniques in AASs, particularly focusing on FL and MARL. Specifically, the survey explores the fundamental characteristics, communication models, and dynamic behaviors of various autonomous agents, including UAVs, UGVs, USVs, and UUVs. Then we provide an overview of the basic ML models used in AASs, discuss their limitations for efficient application, and then highlight the importance

of DML in overcoming these challenges, outlining the essential requirements and metrics for its successful deployment. Furthermore, by categorizing the discussion into two major sections, this work systematically reviews the use cases and key techniques of FL and MARL in enhancing the capabilities of AASs. By examining primary use cases such as cooperative perception, localization and navigation, control, communication, SLAM, and security, the survey highlights the versatility of FL in enabling efficient and secure collaboration among agents. The key techniques to facilitate FL in supporting AASs are also discussed, including aggregation mechanisms, communication compression, resource management strategies, data enhancement methods, and security enhancement approaches. Similarly, the survey explores the application of MARL in AASs across various scenarios, including cooperative communication, navigation, search, formation control, task scheduling, target hunting, and confrontation. Key techniques identified for MARL in AASs encompass foundational learning frameworks, knowledge reuse, sample utilization, task simplification, and communication strategies, which collectively enhance the adaptability and decision-making capabilities of AASs in real-world environments. Finally, this survey outlines several potential research directions to inspire researchers. By systematically presenting the characteristics of autonomous agents, as well as the use cases and key techniques of how FL and MARL can be effectively applied to enhance the capabilities of AASs, this survey provides valuable insights for future research and practical implementation in the field of AASs.

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